

The image shows a well-lit industrial networking lab. In the foreground, there are wooden-topped workbenches with black frames and multiple rows of silver and black drawers. A black ergonomic chair is positioned in front of one of the benches. In the background, several racks of network equipment, including switches and patch panels, are visible. A projector is mounted on the ceiling, and the room has a clean, professional appearance.

INDUSTRIAL NETWORKING MENG 4010

CONTENTS

- 2. Network Communications based on ISO/OSI Model
 - 2.1. Network Communication
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 - 2.4. Encapsulation Process
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 - 2.6. Clients and Servers
 - 2.7. Network Media
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 - 2.9. Two Computers Communicating on a LAN

2.1 Network Communication

Circuit Switching

Circuit switching is a communication method that creates a dedicated path between a sender and a receiver.

- The path is established before data transmission begins.
- The communication resources remain reserved for that connection until the communication ends.
- A simple analogy is a private lane on a highway reserved for one user.

Three Main Steps

1. Establishment

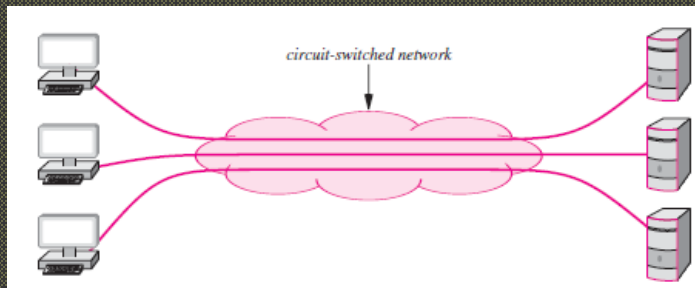
The communication path is set up. Similar to dialing a telephone number and waiting for the call to connect.

2. Data Transfer

Data is transmitted through the established path. The reserved resources are used by that connection.

3. Termination

The connection is disconnected. The reserved resources are released for other users.



Main Characteristics

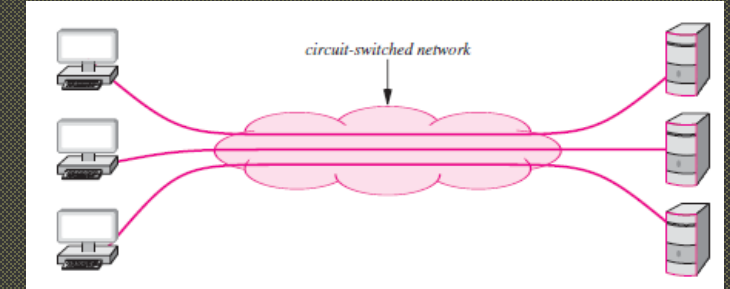
- Provides a point-to-point connection between two endpoints.
- Resources are reserved for the duration of the communication.
- Provides predictable bandwidth and delay.
- Communication is less affected by other network traffic.
- Requires connection setup before data can be transmitted.
- Reserved resources may be unused during periods when no data is being sent.

2.1 Network Communication

Circuit Switching

Examples

- Traditional telephone systems
 - A circuit is established between two callers.
 - The circuit remains active for the duration of the call.
- Dial-up Internet
 - Uses the telephone network to create a circuit-switched connection between the user's modem and the ISP.
- Optical Circuit Switching
 - Uses dedicated optical paths or wavelengths.
 - Can be used in some high-speed communication systems.



Advantages

- Predictable communication performance.
- Dedicated bandwidth.
- Consistent transmission delay.
- Good for continuous communication such as traditional voice calls.

Disadvantages

- Requires time to establish the connection.
- Reserved bandwidth can be wasted when no data is being transmitted.
- Less efficient for bursty computer data.
- A circuit must be available before communication can begin.

Multiplexing in Circuit Switching

- Multiple circuits can share the same physical communication medium.
- Common multiplexing methods include:
 - FDM — Frequency Division Multiplexing
 - Divides the channel into different frequency bands.
 - Each connection uses a different frequency.
 - Multiple users can communicate at the same time.
 - TDM — Time Division Multiplexing
 - Divides communication time into fixed time slots.
 - Each connection is assigned a time slot.
 - Users take turns using the communication channel.

2.1 Network Communication

Packet Switching

- Packet switching is the main communication method used in modern computer networks, including the Internet.
- A large message is divided into smaller units called packets.
- Each packet usually contains:
 - Part of the original data
 - Source address
 - Destination address
 - Control information

How Packet Switching Works

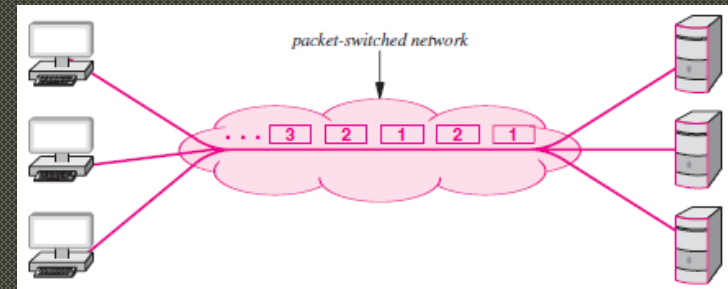
- The sender divides the message into packets.
- Each packet is sent through the network.
- Routers/switches forward packets toward the destination.
- Packets from many users can share the same network links.
- Packets may take different routes through the network.
- At the destination, the data is put back together when required.

Main Properties

- Shared communication
 - Many devices use the same network infrastructure.
- Asynchronous communication
 - Devices send data whenever they need to.
 - There is no fixed transmission schedule for each user.
- No dedicated end-to-end circuit
 - Unlike circuit switching, a permanent path is not reserved for each communication.
- Efficient use of bandwidth
 - Network capacity is used only when devices have data to send.
- Variable performance
 - Performance depends on how busy the network is.

Statistical Multiplexing

- Many users share the same communication channel.
- Bandwidth is assigned based on traffic demand.
- A user does not permanently reserve the communication channel.
- Packets from different users can be interleaved on the same link.
- If one user is not transmitting, other users can use the available bandwidth.



2.1 Network Communication

Packet Switching

Possible Network Problems

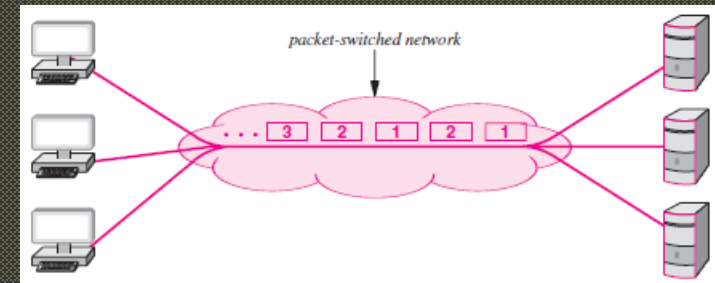
- When many devices send data at the same time:
 - Congestion may occur.
 - Packets may wait in queues.
 - Delay (latency) may increase.
 - Jitter may occur.
 - Some packets may be lost.

Advantages of Packet Switching

- Efficient
 - Many users share the same communication links.
- Cost-effective
 - Dedicated communication paths are not required for every pair of devices.
- Flexible
 - Supports communication between many different devices.
- Scalable
 - Works well for very large networks.
- Reliable and resilient
 - Packets may use alternative routes if another route is available.

Examples

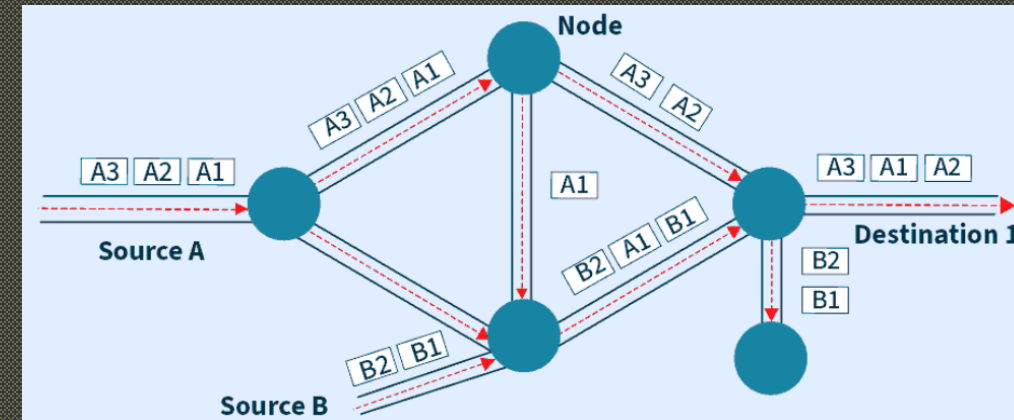
- Internet
 - Web browsing
 - Email
 - Cloud applications
 - Video streaming
- Ethernet networks
 - Computers
 - PLCs
 - HMIs
 - Servers
 - Industrial devices
- Wi-Fi
 - Multiple wireless devices share the same radio channel.
- Cellular networks
 - Many mobile devices share network resources.



2.1 Network Communication

Packet Switching - Datagrams Services

- Connectionless communication
- No path setup before sending
- Message is divided into datagrams
- Each datagram is handled independently
- Datagrams may:
 - Take different routes
 - Arrive out of order
 - Have different delays
 - Be lost
- Higher-layer protocols such as TCP can provide ordering and reliability
- Advantages:
 - Simple
 - Flexible
 - Scalable
 - Efficient use of shared network resources
- Example: Internet Protocol (IP)



<https://www.geeksforgeeks.org/computer-networks/difference-between-datagram-switching-virtual-circuit/>

Example: Internet Protocol (IP) is the best-known example of datagram service.

- When you send an email, stream a video, or browse a website, your data is split into IP packets (datagrams).
- Each one may travel differently across the Internet, but your device reassembles them to make the final message or video play smoothly.

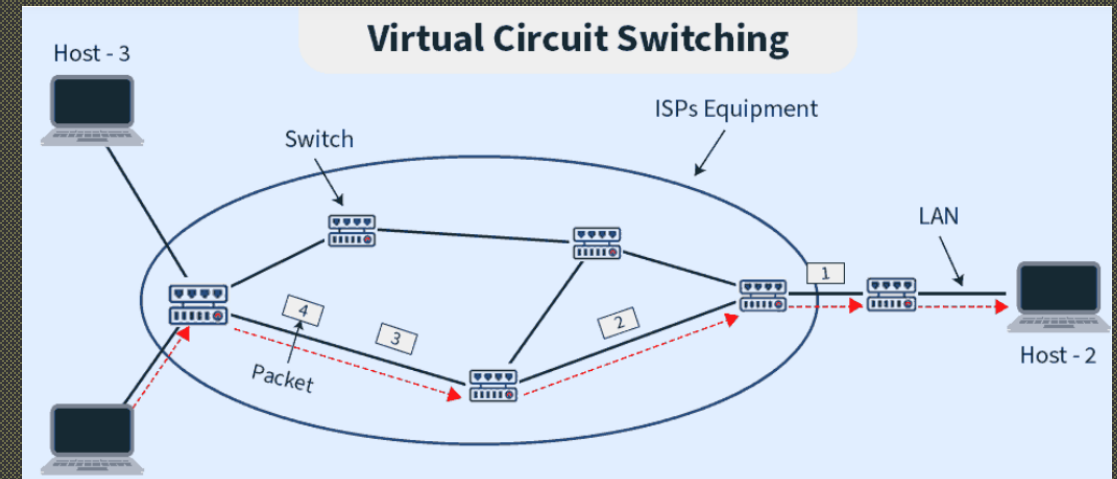
TCP is connection-oriented, but it is not a virtual-circuit switching service.

2.1 Network Communication

Packet Switching - Virtual Circuit (VC) Services

- A logical path is established before data is sent
- Packets normally follow the same logical route
- Communication has 3 phases:
 - Connection setup
 - Data transfer
 - Connection tear-down
- Uses a Virtual Circuit Identifier (VCI) or similar label
- Provides:
 - More predictable forwarding
 - Better packet ordering
 - More consistent performance
- Network resources are still shared
- Examples:
 - ATM
 - Frame Relay

- Datagram
 - No setup
 - Packets handled independently
 - Routes and delays may vary
 - Example: IP
- Virtual Circuit
 - Setup required
 - Packets follow the same logical path
 - More predictable delivery
 - Example: ATM



<https://www.geeksforgeeks.org/computer-networks/difference-between-datagram-switching-virtual-circuit/>

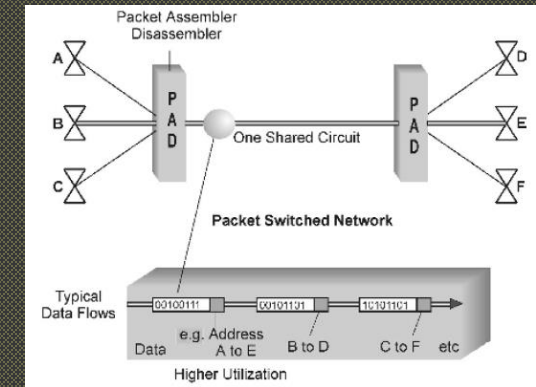
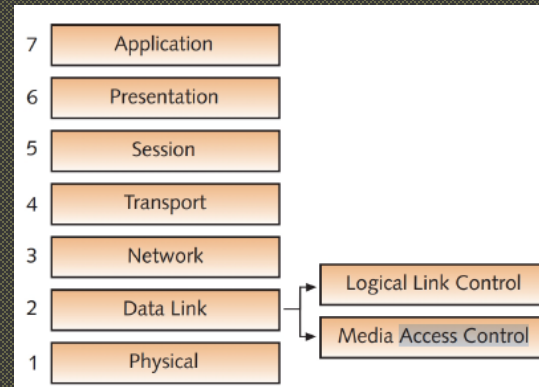
- ATM (Asynchronous Transfer Mode): uses fixed-size cells and VC to provide fast, predictable data transmission
 - Telecom networks, voice/video/data.
- Frame Relay: uses VC to efficiently transmit variable-size data frames over shared wide-area networks
 - Connection of company branch offices to a central headquarters over a WAN.

2.1 Network Communication

IEEE 802 Extensions to the OSI Reference Model

Purpose of IEEE 802 Standards

- IEEE 802 standards define rules mainly for LAN and MAN networks.
- They focus primarily on:
 - Layer 1 – Physical Layer
 - Layer 2 – Data Link Layer
- They standardize how devices:
 - Connect to the network medium
 - Share the communication medium
 - Identify devices on the local network
 - Exchange frames



IEEE 802 divides the Data Link Layer into two sublayers:

- LLC – Logical Link Control
- MAC – Media Access Control

Role of the Physical Layer

- Defines how bits are physically transmitted.
- Specifies items such as:
 - Cables
 - Connectors
 - Electrical signals
 - Optical signals
 - Radio signals
 - Data rates
- Provides the actual physical connection between devices.

Easy Way to Remember

- Physical Layer → How bits physically travel
- MAC Sublayer → Who can use the medium + MAC addressing
- LLC Sublayer → Interface between Data Link Layer and upper layers

2.1 Network Communication

IEEE 802 Extensions to the OSI Reference Model

Logical Link Control (LLC)

- Associated with IEEE 802.2.
- Provides the interface between:
 - Data Link Layer
 - Network Layer
- Helps identify which upper-layer protocol should receive the data.
- Supports multiplexing/demultiplexing of different network-layer protocols.
- Historically could support:
 - Flow control
 - Error control
- In modern Ethernet networks, many of these functions are usually handled by other protocols or layers.

Simple idea

- LLC = communication between Layer 2 and the upper layers.

Media Access Control (MAC)

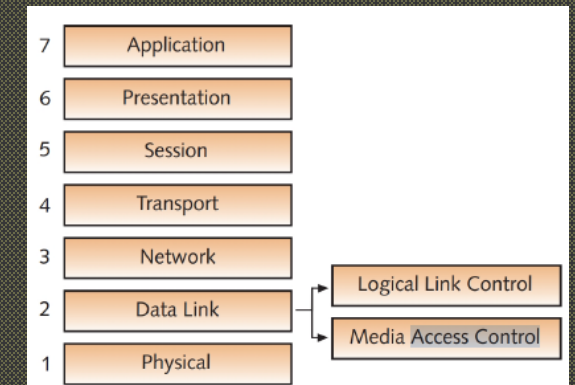
- Controls how devices access the transmission medium.
- Defines rules for sharing the network medium.
- Responsible for MAC addressing.
- Works closely with:
 - Network Interface Card (NIC)
 - Physical Layer
- Helps determine:
 - Who can transmit
 - When transmission can occur
 - How frames are addressed on the LAN

Examples of IEEE 802 technologies

- Ethernet – IEEE 802.3
- Wi-Fi – IEEE 802.11
- Older technologies:
 - Token Ring
 - FDDI - Fiber Distributed Data Interface

Simple idea

- MAC = controls access to the medium and handles local hardware addressing.



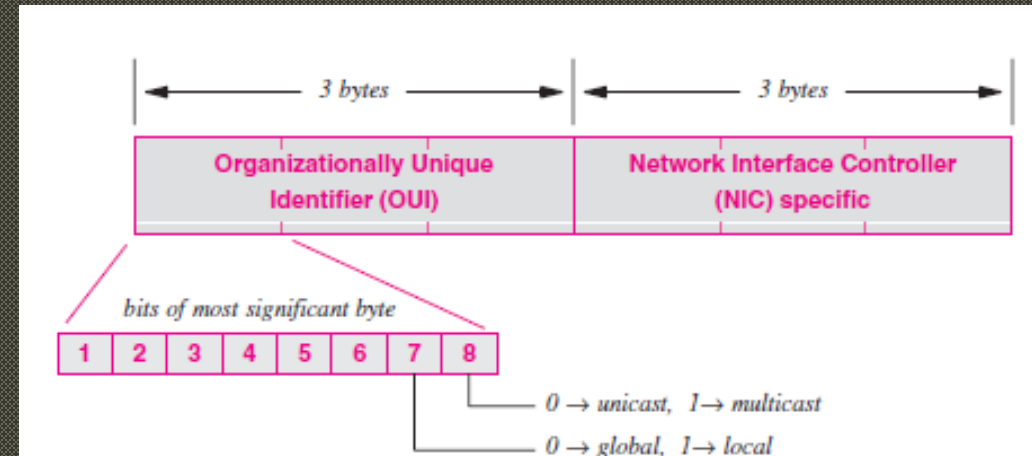
2.1 Network Communication

MAC Address

- A MAC (Media Access Control) address identifies a network interface at the Data Link Layer (Layer 2).
- In Ethernet, MAC addresses are used to deliver frames to the correct device on the local network.
- A typical MAC address is 48 bits (6 bytes) long.
- It is usually written as six hexadecimal pairs:
 - Example: 00:1A:2B:3C:4D:5E

MAC Address Structure

- A traditional 48-bit MAC address can be viewed as two main parts:
 - OUI – Organizationally Unique Identifier
 - First 3 bytes
 - Identifies the manufacturer/vendor address block
 - Device-specific portion
 - Last 3 bytes
 - Assigned to a particular network interface



2.1 Network Communication

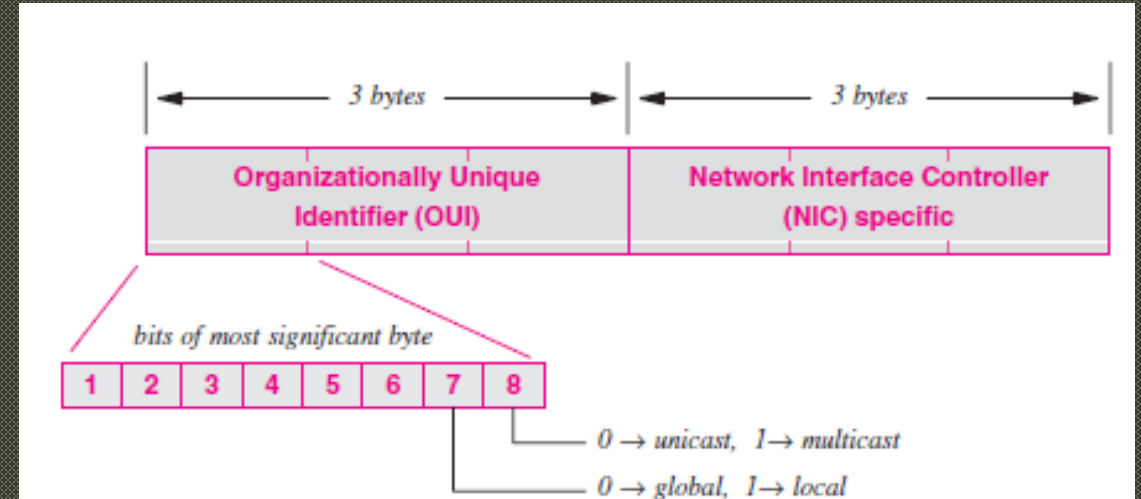
MAC Address

Special Bits

- Two important bits appear in the first byte:
- I/G bit – Individual/Group
 - 0 = Unicast → one device
 - 1 = Multicast → group of devices
- U/L bit – Universal/Local
 - 0 = Universally administered address
 - 1 = Locally administered address

How Uniqueness Is Managed

- IEEE assigns address blocks to manufacturers.
- Manufacturers assign addresses to their network interfaces.
- This is intended to make factory-assigned MAC addresses globally unique.
- However, MAC addresses can also be:
 - Changed by software
 - Locally assigned
 - Randomized by modern devices



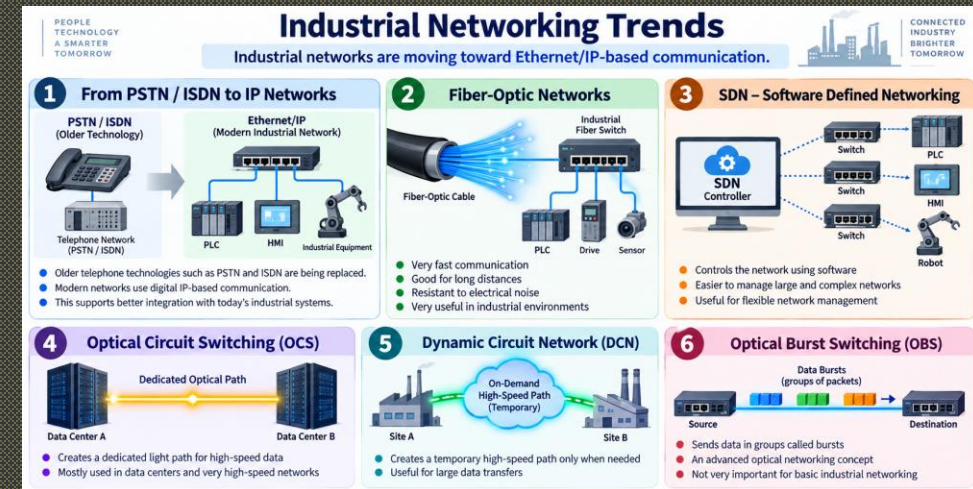
- MAC address = Layer 2 address of a network interface
- 48 bits = 6 bytes
- First part → vendor/address block
- Second part → individual interface
- Unicast → one device
- Multicast → group of devices

2.1 Network Communication

Industry Updates

- Industrial networks are moving toward Ethernet/IP-based communication.
- Older telephone technologies such as PSTN and ISDN are being replaced by modern digital IP networks.
- Fiber-optic networks
 - Used for very fast communication.
 - Good for long distances.
 - Resistant to electrical noise.
 - Very useful in industrial environments.
- SDN – Software Defined Networking
 - Makes networks easier to control using software.
 - Helps manage large and complex networks.
- Optical Circuit Switching (OCS)
 - Creates a dedicated light path for high-speed data.
 - Mostly used in data centers and very high-speed networks.

- **PSTN: Public Switched Telephone Network** - traditional analog phone network using copper wires.
- **ISDN: Integrated Services Digital Network** - digital tech for faster voice/data over one line.

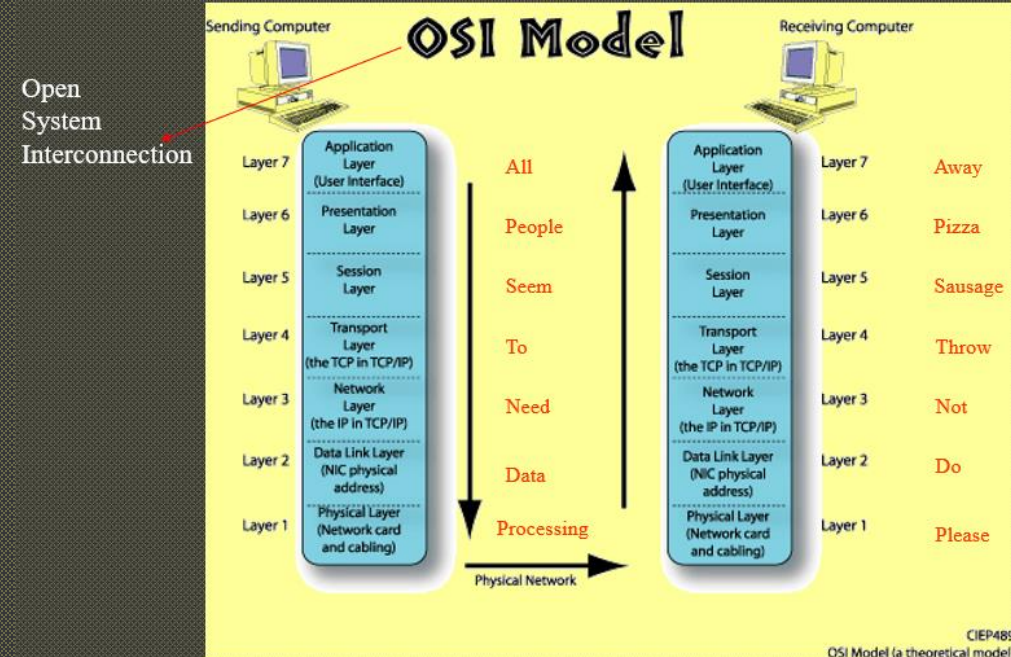


- **Dynamic Circuit Network (DCN)**
 - Creates a temporary high-speed path only when needed.
 - Useful for large data transfers.
- **Optical Burst Switching (OBS)**
 - Sends data in groups called bursts.
 - Mainly an advanced optical networking concept.
 - Not very important for basic industrial networking

2.2 OSI REFERENCE MODEL

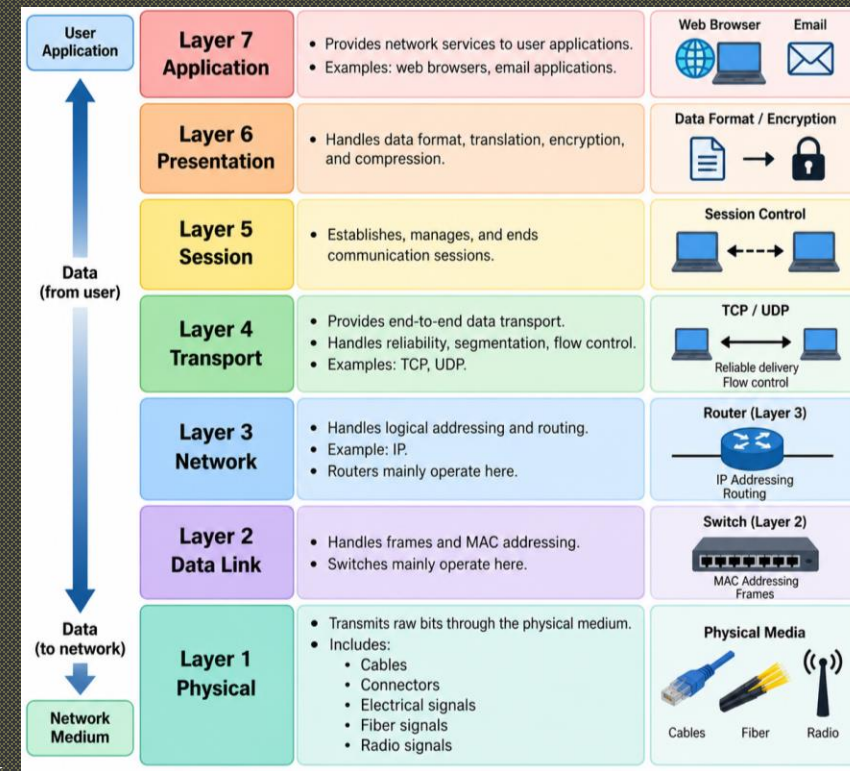
What is the OSI Model?

- Developed by ISO – International Organization for Standardization.
- OSI stands for Open Systems Interconnection.
- Provides a common framework for understanding network communication.
- Divides networking functions into 7 layers.
- Helps:
 - Network designers
 - Protocol developers
 - Students
 - Technicians
- Very useful for network troubleshooting.
- The OSI model is a reference model, not a specific protocol.



2.2 OSI REFERENCE MODEL

- Layer 7 – Application
 - Provides network services to user applications.
 - Examples: web browsers, email applications.
- Layer 6 – Presentation
 - Handles data format, translation, encryption, and compression.
- Layer 5 – Session
 - Establishes, manages, and ends communication sessions.
- Layer 4 – Transport
 - Provides end-to-end data transport.
 - Handles reliability, segmentation, flow control.
 - Examples: TCP, UDP.
- Layer 3 – Network
 - Handles logical addressing and routing.
 - Example: IP.
 - Routers mainly operate here.



- Layer 2 – Data Link
 - Handles frames and MAC addressing.
 - Switches mainly operate here.
- Layer 1 – Physical
 - Transmits raw bits through the physical medium.
 - Includes:
 - Cables
 - Connectors
 - Electrical signals
 - Fiber signals
 - Radio signals

2.2 OSI REFERENCE MODEL

Role of a Reference Model

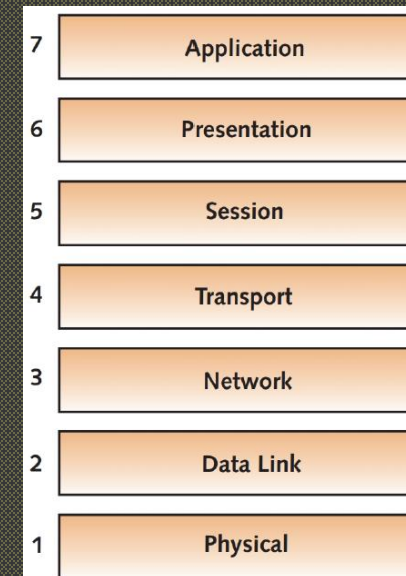
- Each task = a layer:
 - Each layer performs its own function and provides services to adjacent layers.
- Advantages of layering:
 - Breaks a big problem into smaller, manageable tasks.
 - Independence: One step can change without affecting the rest.

Example of Independence in Layers

- Postal analogy: Mail carrier switches from walking to driving →
 - Only delivery method changes.
 - Addressing, sorting, and reading steps remain the same.
- Networking example:
 - Upgrading from 100 Mbps Ethernet → 1 Gbps Ethernet:
 - Only NICs and switches change.
 - Applications/protocols remain unchanged.
 - IPv4 → IPv6 upgrade:
 - Changes at Network layer only in a simple way (OS, DNS, or security rules may need to change also)

Analogy: Sending a Letter

- Write the letter → message creation.
- Put in an envelope → packaging.
- Add addresses → sender + recipient info.
- Add a stamp → authorization to send.
- Drop in mailbox → sender's job done.
- Mail carrier picks it up → delivery starts.
- Sorting by zip code (ON) → gets sent to right region.
- Sorting by zip code (TO) → gets sent to right neighborhood.
- Mail carrier delivers to house → final delivery step.
- Recipient opens and reads → message successfully received.



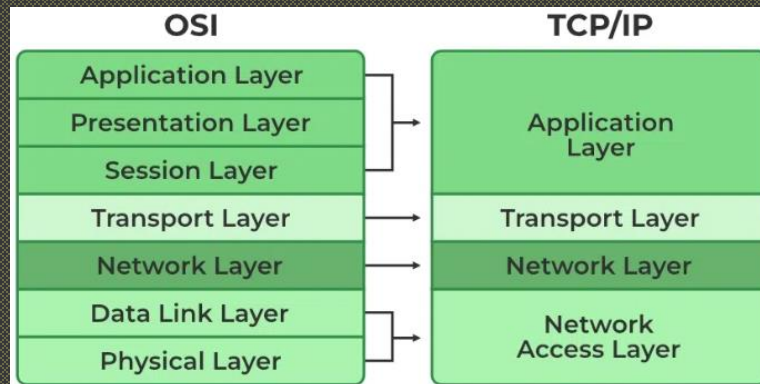
2.2 OSI REFERENCE MODEL

OSI vs TCP/IP

- OSI
 - 7-layer reference model.
 - Mainly used for teaching, design, and troubleshooting.
- TCP/IP
 - Practical protocol suite used on modern networks and the Internet.
 - Usually represented with fewer layers.

Typical Mapping

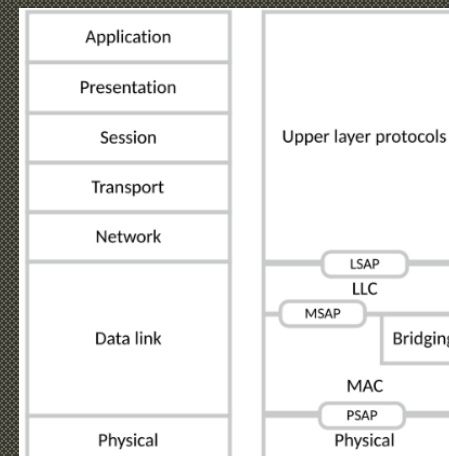
- OSI Layers 7, 6, 5 → TCP/IP Application Layer
- OSI Layer 4 → TCP/IP Transport Layer
- OSI Layer 3 → TCP/IP Internet/Network Layer
- OSI Layers 2, 1 → TCP/IP Network Access Layer



<https://www.geeksforgeeks.org/computer-networks/tcp-ip-model/>

OSI vs IEEE 802

- OSI
 - General conceptual networking model.
- IEEE 802
 - Defines actual LAN/MAN standards.
 - Mainly focuses on:
 - Layer 1 – Physical
 - Layer 2 – Data Link
- Examples:
 - IEEE 802.3 – Ethernet
 - IEEE 802.11 – Wi-Fi



LSAP – Link Service Access Point
MSAP – MAC Service Access Point
PSAP – Physical Service Access Point

2.2 OSI REFERENCE MODEL

Comparison with the TCP/IP Model

1. Application Layer

- OSI: Uses three separate layers:
 - Layer 7 – Application
 - Layer 6 – Presentation
 - Layer 5 – Session
- TCP/IP: Combines these into one Application Layer.
- Examples:
 - HTTP/HTTPS
 - FTP
 - DNS
 - SMTP
 - DHCP
- Functions may include:
 - User/application network services
 - Data formatting
 - Encryption
 - Session management

Simple mapping:

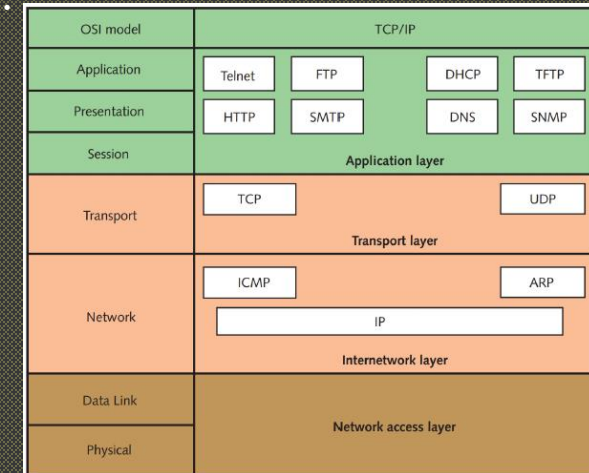
- OSI 7 + 6 + 5 → TCP/IP Application

2. Transport Layer

- Both models have a Transport Layer.
- Provides communication between applications on end devices.
- Main TCP/IP transport protocols:
 - TCP
 - Connection-oriented
 - Reliable
 - Ordered delivery
 - Retransmits lost data
 - UDP
 - Connectionless
 - Lower overhead
 - No guaranteed delivery or ordering
- Examples:
 - TCP → web traffic, file transfer, email
 - UDP → streaming, VoIP, DNS, some industrial protocols

Simple mapping:

- OSI Layer 4 → TCP/IP Transport



2.2 OSI REFERENCE MODEL

Comparison with the TCP/IP Model

3. Network Layer

- OSI: Called the Network Layer.
- TCP/IP: Usually called the Internet Layer.
- Responsible for:
 - IP addressing
 - Routing
 - Moving packets between networks
- Main protocol:
 - IP – IPv4 / IPv6
- Routers mainly operate here.

Simple mapping:

- OSI Layer 3 → TCP/IP Internet Layer

OSI model	TCP/IP			
Application	Telnet	FTP	DHCP	TFTP
Presentation	HTTP	SMTP	DNS	SNMP
Session	Application layer			
Transport	TCP		UDP	
	Transport layer			
Network	ICMP			ARP
	IP			
	Internetwork layer			
Data Link	Network access layer			
Physical				

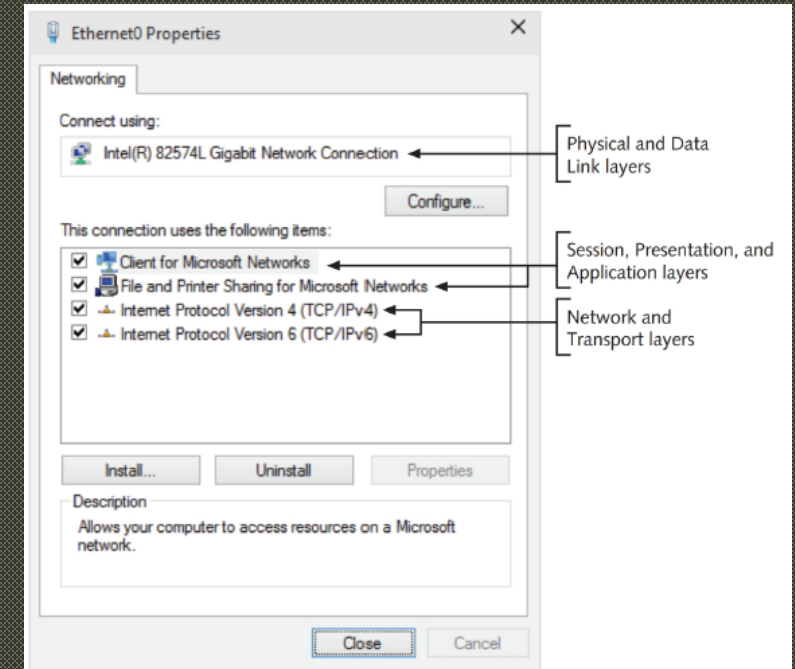
4. Data Link + Physical Layers

OSI separates them:

- Layer 2 – Data Link
 - Frames
 - MAC addresses
 - Ethernet switching
- Layer 1 – Physical
 - Cables
 - Connectors
 - Electrical signals
 - Fiber signals
 - Radio signals

TCP/IP

- Usually combines these functions into the Network Access / Link Layer.
- Simple mapping:
- OSI Layers 2 + 1 → TCP/IP Network Access Layer



2.2 OSI REFERENCE MODEL

Layers of the Model

1. Physical Layer (Layer 1)

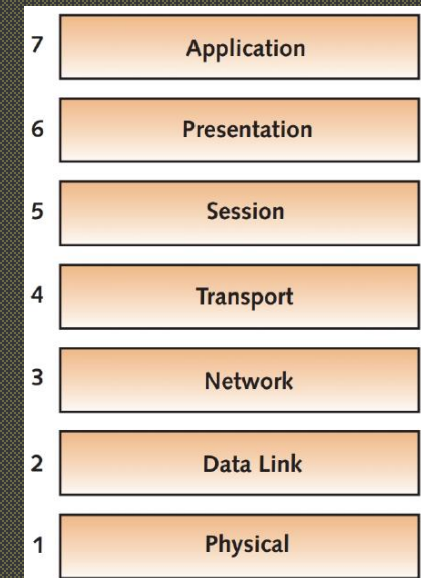
- Concerned with the physical medium: cables, signals, connectors, and voltages.
- Deals with transmitting raw bits (0s and 1s) over the medium.
- Examples:
 - Copper cables, fiber optics, Wi-Fi radio waves.
 - Hubs and NICs (at the hardware signal level).

2. Data Link Layer (Layer 2)

- Provides framing and error detection between devices on the same local network/link.
- Handles framing, MAC addresses, and error detection (CRC).
- Examples:
 - Ethernet (MAC addresses), Wi-Fi (802.11).
 - Devices: Switches, bridges, NICs (logical layer).

3. Network Layer (Layer 3)

- Handles logical addressing and routing of data across multiple networks.
- selects a path based on routing information and routing metrics.
- Examples:
 - IP (IPv4/IPv6).
 - Devices: Routers, Layer 3 switches.



2.2 OSI REFERENCE MODEL

Layers of the Model

4. Transport Layer (Layer 4)

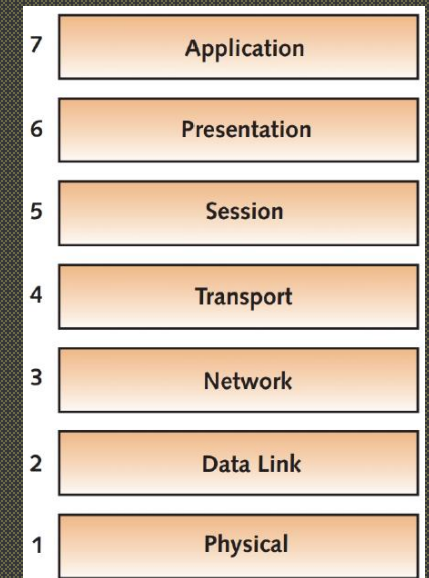
- Provides end-to-end transport between applications.
- Breaks data into segments and ensure that data is delivered in the correct order.
- Examples:
 - TCP provides reliability, sequencing, retransmission, and flow control.
 - UDP does not provide guaranteed delivery, ordering, or retransmission.
 - Device/software: Firewalls often work at this layer (they inspect layers 3 – 7).

5. Session Layer (Layer 5)

- Manages sessions/connections between applications.
- Establishes, maintains, and ends communication sessions.
- Examples:
 - Remote Desktop Protocol (RDP), NetBIOS sessions.
 - Logging into a server and keeping the session alive.

6. Presentation Layer (Layer 6)

- Translates data into a format applications can understand.
- Handles encryption, compression, and data translation.
- Examples:
 - TLS (HTTPS encryption).
 - JPEG, MP4, GIF file formats.



2.2 OSI REFERENCE MODEL

Layers of the Model

7. Application Layer (Layer 7)

- Closest to the user. Provides network services to applications.
- Interfaces directly with programs.
- Examples:
 - Web browsers (HTTP/HTTPS), email (SMTP), file sharing (FTP).
 - Devices: PCs, smartphones, servers.

Layer Interactions

- Each layer works with the one above and below it.
 - Example: Transport layer relies on Network layer for addressing/routing and serves Session layer for connection management.
- Top layer (Application): interacts with user programs (e.g., Word, File Explorer, Chrome).
- Bottom layer (Physical): deals with signals directly.

Layer #	Layer Name	Main Function	Protocols / Standards	Devices	Examples (Everyday Use)
7	Application	Provides programs with access to network services	HTTP, HTTPS, SMTP, FTP, DNS	PCs, smartphones, servers	Using a web browser to load a webpage; sending an email
6	Presentation	Translates, encrypts, compresses data	SSL/TLS, JPEG, GIF, MPEG, ASCII, EBCDIC	PCs/servers (software-based)	Watching a YouTube video (MPEG); SSL encryption in HTTPS
5	Session	Manages sessions (start, maintain, end connections)	NetBIOS, RPC	Gateways, application servers	Remote Desktop Protocol (RDP) session; maintaining a VoIP call
4	Transport	End-to-end delivery, segmentation, reliability, flow control	TCP, UDP	Firewalls, load balancers	TCP ensures a file downloads fully; UDP used in live streaming
3	Network	Logical addressing, routing, path determination	IPv4, IPv6, ICMP	Routers, Layer 3 switches	Sending data across the internet using an IP address
2	Data Link	Error detection, framing, physical addressing (MAC)	Ethernet (802.3), Wi-Fi (802.11)	Switches, bridges, NICs	Wi-Fi connection at home; Ethernet LAN in offices
1	Physical	Transmission of raw bits over a medium	IEEE 802.3 (physical specs), RS-232, DSL	Cables, hubs, repeaters, NICs	Fiber optic cable transmitting light signals; Cat6 cable carrying Ethernet

2.2 OSI REFERENCE MODEL

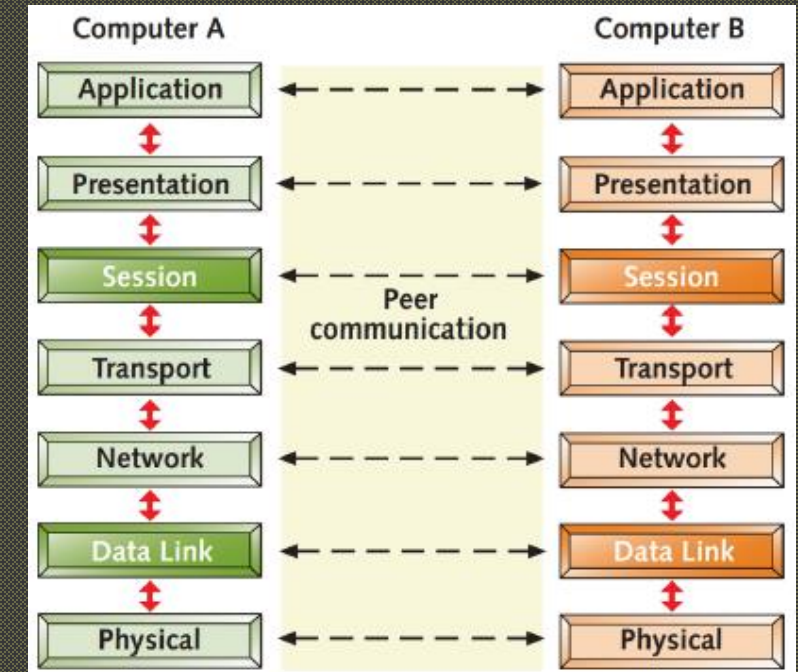
How Communication Works

1. Layered Services

- Each OSI layer:
 - Performs a specific function.
 - Uses services from the layer below.
 - Provides services to the layer above.
- Example:
 - Physical Layer transmits bits for the Data Link Layer.
 - Network Layer provides routing services for the Transport Layer.

2. Peer-to-Peer Communication

- The same layers on two devices are called peer layers.
- For example:
 - Transport Layer on Computer A logically communicates with the Transport Layer on Computer B.
 - Network Layer communicates logically with the Network Layer.
- This communication is logical, not a direct physical connection.
- Each peer layer follows the same protocol rules to understand the information it receives.



- Vertical communication = between layers inside one computer.
- Horizontal/peer communication = logical communication between the same layers on two computers.

2.2 OSI REFERENCE MODEL

How Communication Works

3. Protocol Stack Processing

Sending Computer – Encapsulation

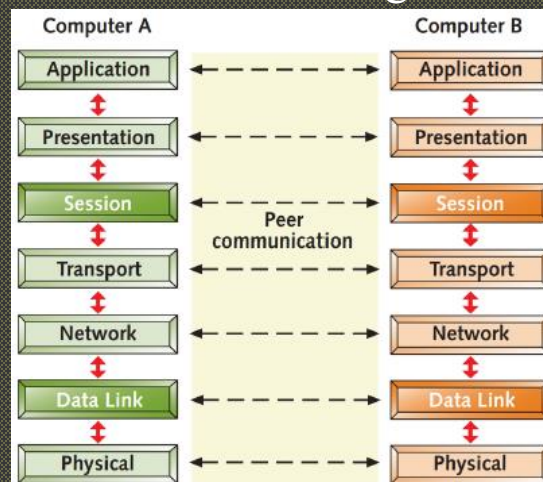
- Data starts at the Application Layer.
- It moves down through the OSI layers.
- As it moves downward, layers add control information such as:
 - Addresses
 - Protocol information
 - Sequence information
 - Error-checking information
- This process is called encapsulation.
- The Physical Layer finally sends the information as bits/signals across the network.

Typical PDU Names

- Layers 7–5 → Data
- Layer 4 → Segment for TCP / Datagram for UDP
- Layer 3 → Packet
- Layer 2 → Frame
- Layer 1 → Bits

Example

- Original message → data
- Add TCP information → segment
- Add IP information → packet
- Add Ethernet information → frame
- Convert to signals → bits transmitted



2.2 OSI REFERENCE MODEL

How Communication Works

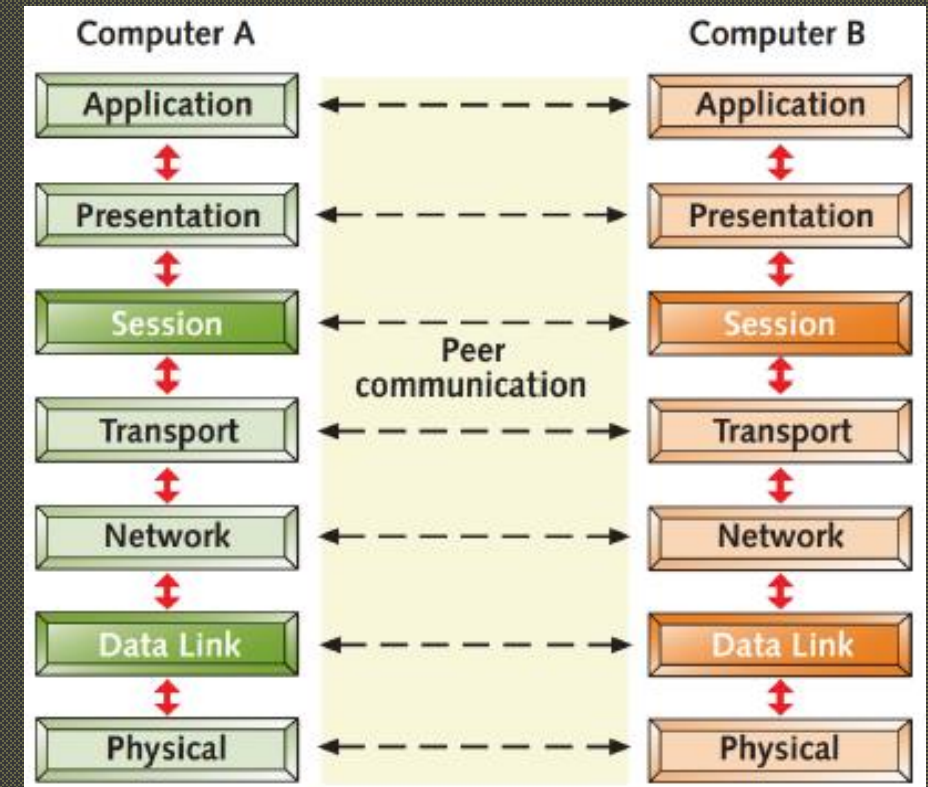
3. Protocol Stack Processing

Receiving Computer – De-Encapsulation

- The receiving computer performs the reverse process.
- Data moves from:
 - Physical Layer → Application Layer
- Each appropriate layer:
 - Reads its control information.
 - Removes the related header/trailer.
 - Passes the remaining information upward.
- This process is called de-encapsulation.
- Eventually, the original application data reaches the receiving program.

Simple way to remember

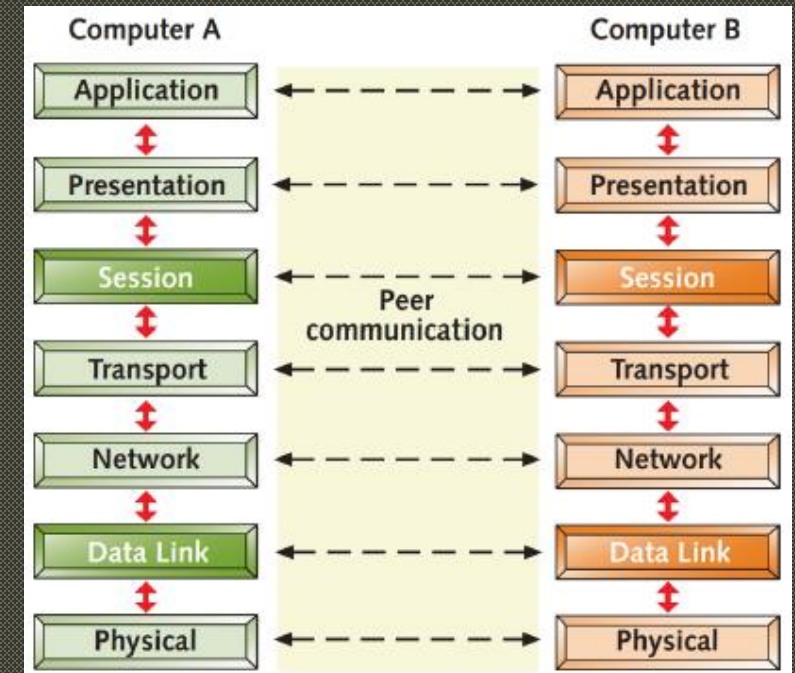
- Sender: Encapsulation → adds information.
- Receiver: De-encapsulation → removes information.



2.2 OSI REFERENCE MODEL

How Communication Works

- Network devices are often described by the highest layer used for their main networking function.
- Layer 1 – Physical
 - Cables
 - Repeaters
 - Hubs
- Layers 1–2
 - NICs generally perform both Physical and Data Link functions.
- Layer 2 – Data Link
 - Ethernet switches
 - Bridges
- Layer 3 – Network
 - Routers
 - Layer 3 switches
- Layers 3–7
 - Firewalls may operate across several layers depending on their capabilities.
- Layer 7
 - Application proxies
 - Application gateways
 - Some advanced firewalls/load balancers



- Proxy = stands in the middle, between a client and a server.
- Gateway = controls entry to an application/service.
- Advanced firewall = inspects and protects application traffic.
- Load balancer = spreads application traffic across multiple servers.

2.2 OSI REFERENCE MODEL

Application Layer

- Provides the interface between user applications and network services.
- Supports services such as:
 - Web browsing
 - Email
 - File sharing
 - Database access
 - Remote access
- The PDU at Layers 5-7 is generally referred to as data.

Client/Server Communication

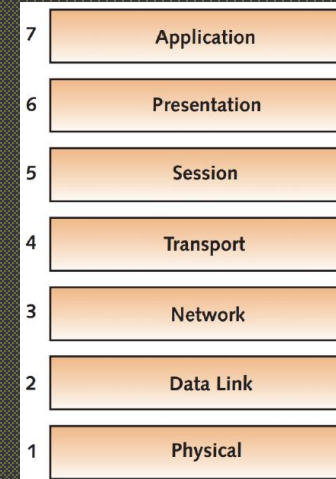
- Many Application-layer services use a client/server model.
- Examples:
 - Web browser ↔ Web server
 - File-sharing client ↔ File server
 - NFS client ↔ NFS server

Possible Application-Layer Problems

- Incorrect application configuration
- Client and server software incompatibility
- Wrong application settings
- Server unavailable
- Authentication problems
- Application protocol errors

Error Handling

- Application-layer software may provide its own error handling.
- Reliability may also come from lower layers.
- Example:
 - HTTP over TCP benefits from TCP reliability.
 - TFTP uses UDP, so TFTP provides its own acknowledgements and retransmissions.



Common Application-Layer Protocols

- HTTP/HTTPS
 - Web browsing
- FTP
 - File transfer
- TFTP
 - Simple file transfer
 - Uses UDP
 - Often used for configuration or boot files
- SMTP
 - Sending email
- Other common examples:
 - DNS
 - DHCP
 - SNMP
 - SSH

Devices/Software Associated with Layer 7

- Application proxies
- Web/application gateways
- Layer 7 load balancers
- Some advanced firewalls
- Application-aware IDS/IPS systems

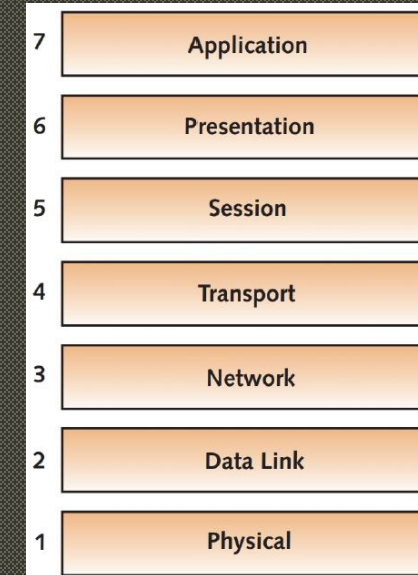
2.2 OSI REFERENCE MODEL

Presentation Layer

- Handles how data is formatted, represented, and interpreted.
- Makes sure data from one system can be understood by another.
- Works between the Application Layer and the lower networking layers.

Main Functions

- Data translation
 - Converts data between different formats or representations.
- Encryption / Decryption
 - Protects data before transmission.
 - Converts encrypted data back to readable form at the destination.
- Compression / Decompression
 - Reduces data size before transmission.
 - Restores the original data after reception.
- Character encoding
 - Helps systems interpret text correctly.
 - Examples: ASCII, Unicode.
- Data-format handling (Examples: JPEG, GIF, PNG, MPEG).



Examples

- Converting text between different character formats.
- Converting line endings such as:
 - Windows → CR/LF and Linux/UNIX → LF
- Encrypting and decrypting secure network data.
- Compressing images, audio, or other data.

2.2 OSI REFERENCE MODEL

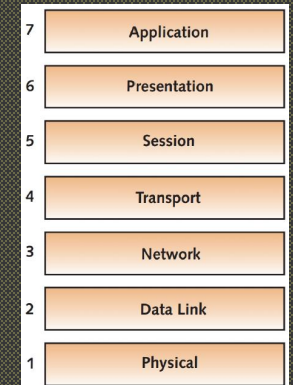
Session Layer

- Establishes, manages, and terminates communication sessions between applications.
- Keeps communication organized while two applications exchange data.
- Main functions
 - Session setup
 - Starts the communication session.
 - Session maintenance
 - Keeps the session active while data is exchanged.
 - Session termination
 - Ends the session when communication is complete.
 - Dialog control
 - Helps coordinate which side communicates and when.
 - Synchronization / checkpointing
 - Can create checkpoints during long communications.
 - If communication is interrupted, it may allow recovery from a checkpoint instead of starting again.
- Example of synchronization
 - During a long file transfer, checkpoints can help resume communication after an interruption.
 - In multimedia communication, synchronization helps keep related data streams coordinated.

The Session Layer is mainly a conceptual OSI function.

In modern TCP/IP networks, Session Layer functions are often built into:

- Applications
- Application protocols
- Operating systems
- Other upper-layer software



2.2 OSI REFERENCE MODEL

Transport Layer

- Provides end-to-end communication between applications.
- Uses port numbers to identify the correct application or service.
- Breaks large amounts of data into smaller units for transmission.
- Reassembles received data at the destination.

Main Functions

- Segmentation
 - Breaks application data into smaller pieces.
- Application identification
 - Uses source and destination port numbers.
- Multiplexing / Demultiplexing
 - Allows multiple applications to communicate at the same time.
 - Directs received data to the correct application.

Data created by the Application, Presentation, and Session layers:

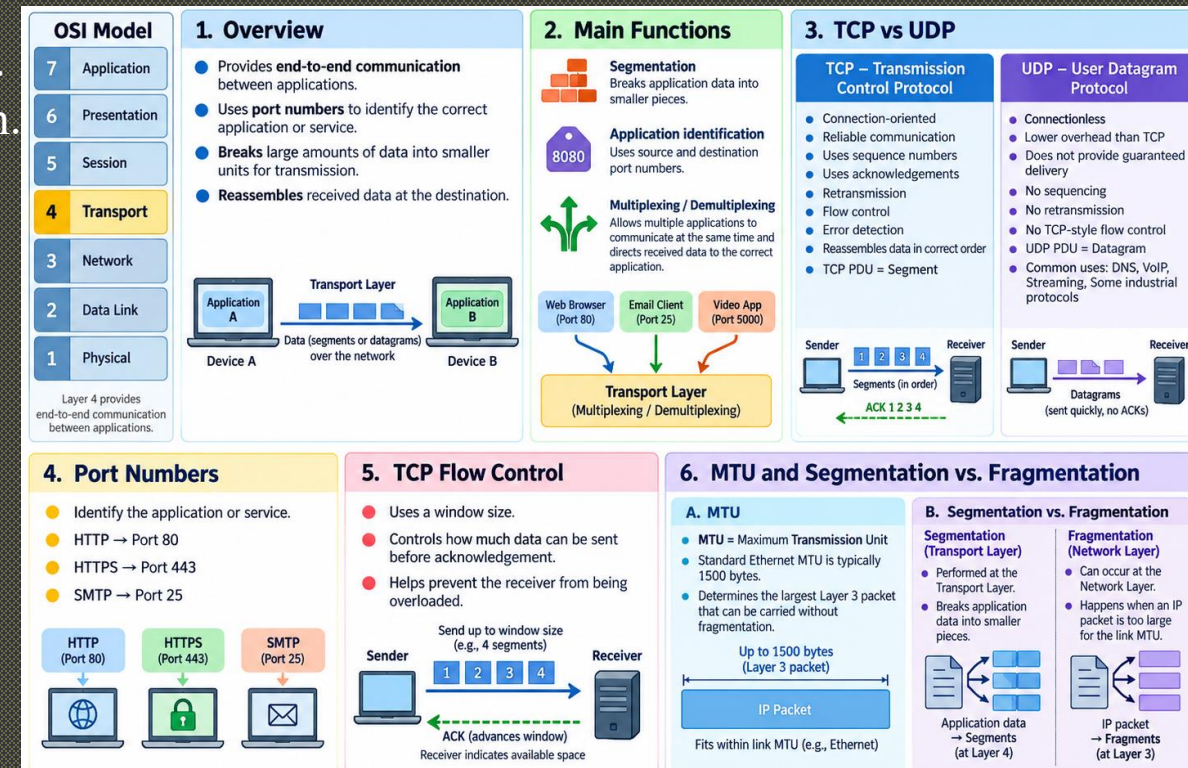
Data data data data data data data data data data
Data data data data data data data data data data
Data data data data data data data data data data

Data is broken into smaller chunks by the Transport layer:

Transport-layer header: Segment 1 Data data data data data data data data data data

Transport-layer header: Segment 2 Data data data data data data data data data data

Transport-layer header: Segment 3 Data data data data data data data data data data



2.2 OSI REFERENCE MODEL

Transport Layer

TCP – Transmission Control Protocol

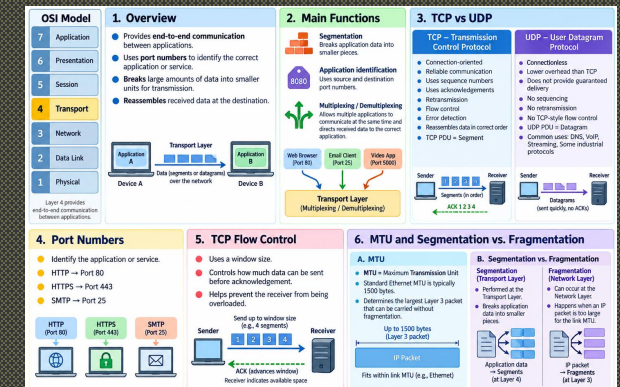
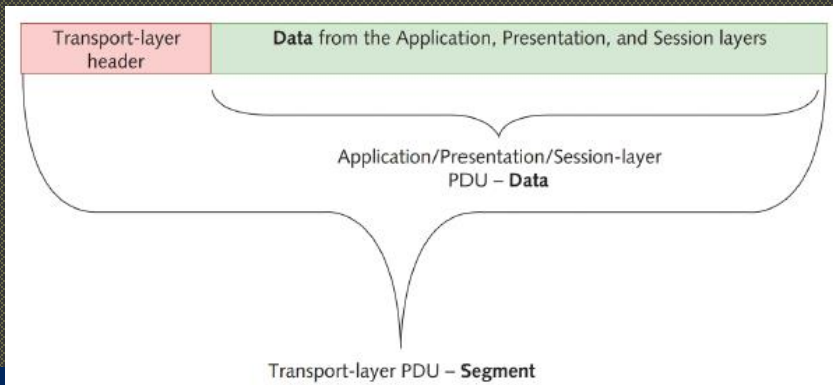
- Connection-oriented.
- Provides reliable communication.
- Uses:
 - Sequence numbers
 - Acknowledgements
 - Retransmission
 - Flow control
 - Error detection
- Ensures data is reassembled in the correct order.
- TCP PDU is called a segment.

UDP – User Datagram Protocol

- Connectionless.
- Lower overhead than TCP.
- Does not provide:
 - Guaranteed delivery
 - Sequencing
 - Retransmission
 - TCP-style flow control
- UDP PDU is called a datagram.
- Commonly used for:
 - DNS
 - VoIP
 - Streaming
 - Some industrial protocols

Port Numbers

- Identify the application or service.
- Examples:
 - HTTP → Port 80
 - HTTPS → Port 443
 - SMTP → Port 25



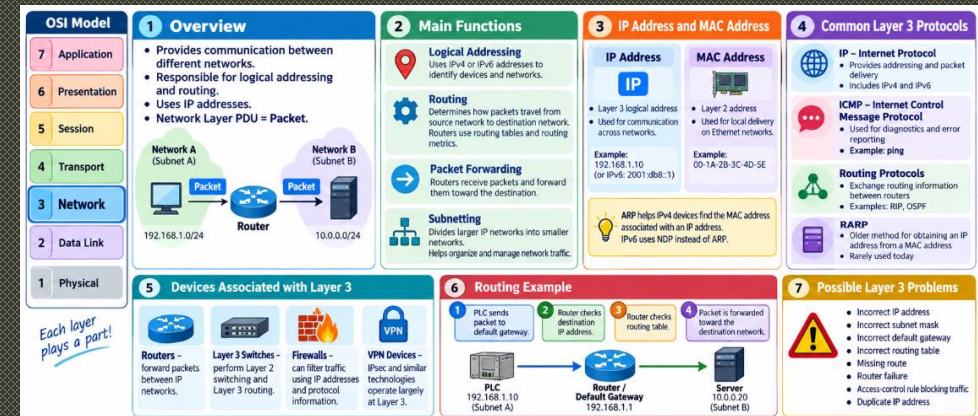
2.2 OSI REFERENCE MODEL

Network Layer

- Provides communication between different networks.
- Responsible for logical addressing and routing.
- Uses IP addresses.
- Network Layer PDU is called a packet.

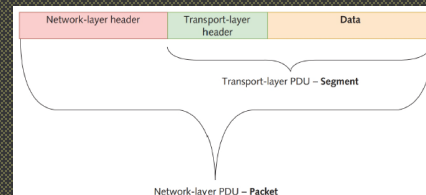
Main Functions

- Logical Addressing
 - Uses IPv4 or IPv6 addresses to identify devices and networks.
- Routing
 - Determines how packets travel from the source network to the destination network.
 - Routers use routing tables and routing metrics.
- Packet Forwarding
 - Routers receive packets and forward them toward their destination.
- Subnetting
 - Divides larger IP networks into smaller networks.
 - Helps organize and manage network traffic.



Routing Example

- A PLC sends data to a server on another subnet.
- The PLC sends the packet to its default gateway.
- The router checks the destination IP address.
- The router checks its routing table.
- The packet is forwarded toward the destination network.



2.2 OSI REFERENCE MODEL

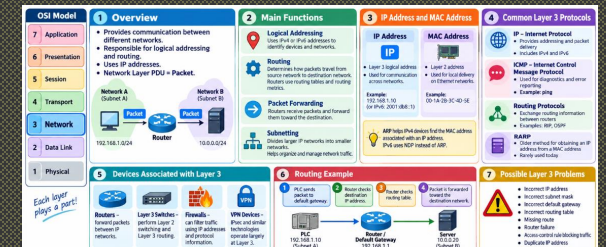
Network Layer

IP Address and MAC Address

- IP address
 - Layer 3 logical address.
 - Used for communication across networks.
 - MAC address
 - Layer 2 address.
 - Used for local delivery on Ethernet networks.
 - ARP
 - Helps IPv4 devices find the MAC address associated with an IP address.
 - IPv6
 - Uses Neighbor Discovery Protocol (NDP) instead of ARP.
- Devices Associated with Layer 3**
- Routers
 - Forward packets between different IP networks.
 - Layer 3 Switches
 - Perform both Layer 2 switching and Layer 3 routing.
 - Firewalls
 - Can filter traffic using IP addresses and protocol information.
 - VPN Devices
 - Some VPN technologies, such as IPsec, operate largely at Layer 3.

Common Layer 3 Protocols

- IP – Internet Protocol
 - Provides addressing and packet delivery.
 - Includes IPv4 and IPv6.
- ICMP – Internet Control Message Protocol
 - Used for network diagnostics and error reporting.
 - Example: ping
- Routing Protocols
 - Exchange routing information between routers.
 - Examples:
 - RIP
 - OSPF
- RARP
 - Older method for obtaining an IP address from a MAC address.
 - Rarely used today.



2.2 OSI REFERENCE MODEL

Data Link Layer

- Provides node-to-node communication over a local network link.
- Works with frames.
- Acts between:
 - Layer 3 – Network and Layer 1 – Physical
- Data Link Layer PDU is called a frame.
- Handles:
 - MAC addressing
 - Frame creation and delivery
 - Media access
 - Error detection

Main Functions

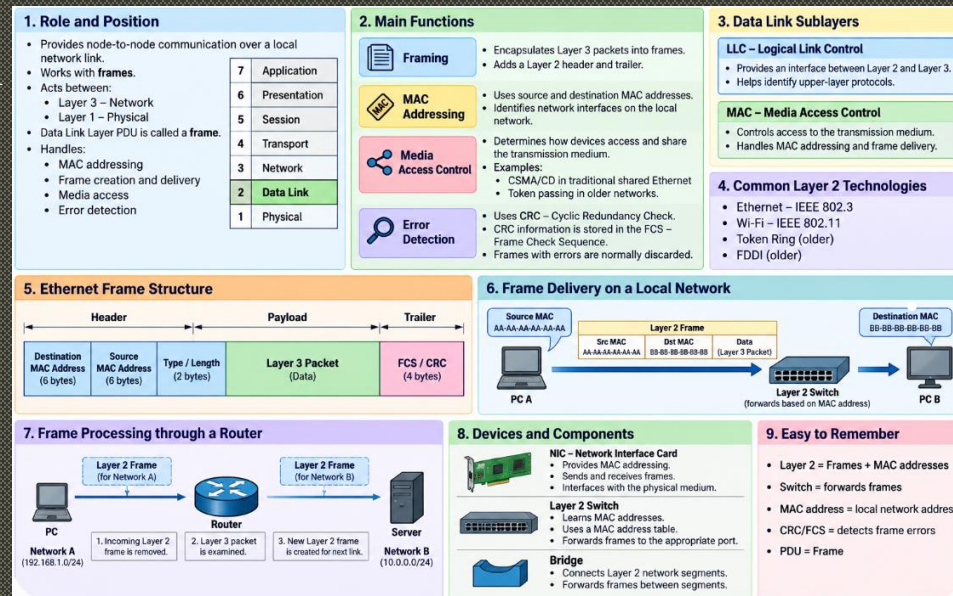
- Framing
 - Encapsulates Layer 3 packets into frames.
 - Adds a Layer 2 header and trailer.
- MAC Addressing
 - Uses source and destination MAC addresses.
 - Identifies network interfaces on the local network.
- Media Access Control
 - Determines how devices access and share the transmission medium.

Examples:

- CSMA/CD in traditional shared Ethernet
- Token passing in older networks

Error Detection

- Uses CRC – Cyclic Redundancy Check.
- CRC information is stored in the FCS – Frame Check Sequence.
- Frames containing detected errors are normally discarded.



2.2 OSI REFERENCE MODEL

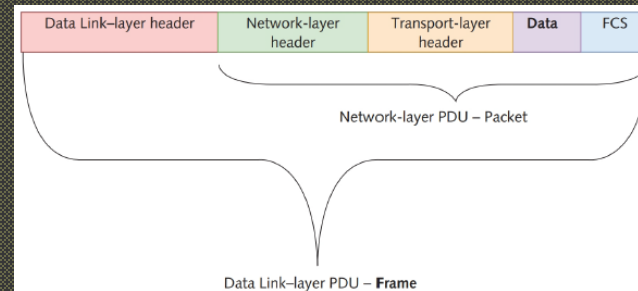
Data Link Layer

Ethernet Frame Structure

- Header
 - Destination MAC address
 - Source MAC address
 - Control/type information
- Payload
 - Contains the Layer 3 packet.
- Trailer
 - Contains the FCS/CRC for error detection.

Frame Delivery

- Source MAC address
 - Identifies the interface sending the frame on the local link.
- Destination MAC address
 - Identifies the receiving interface or next-hop device.
- When a packet passes through a router:
 - The incoming Layer 2 frame is removed.
 - The Layer 3 packet is examined.
 - A new Layer 2 frame is created for the next network link.



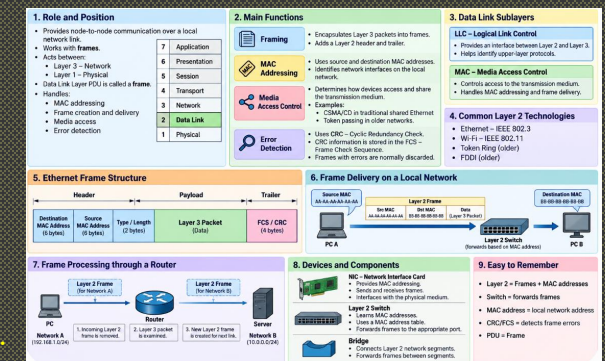
Data Link Sublayers

LLC – Logical Link Control

- Provides an interface between Layer 2 and Layer 3.
- Helps identify upper-layer protocols.

MAC – Media Access Control

- Controls access to the transmission medium.
- Handles MAC addressing and frame delivery.



Devices and Components

- **NIC – Network Interface Card**
 - Provides MAC addressing.
 - Sends and receives frames.
 - Interfaces with the physical network medium.
- **Layer 2 Switch**
 - Learns MAC addresses.
 - Uses a MAC address table.
 - Forwards frames to the appropriate switch port.
- **Bridge**
 - Connects Layer 2 network segments and forwards frames.

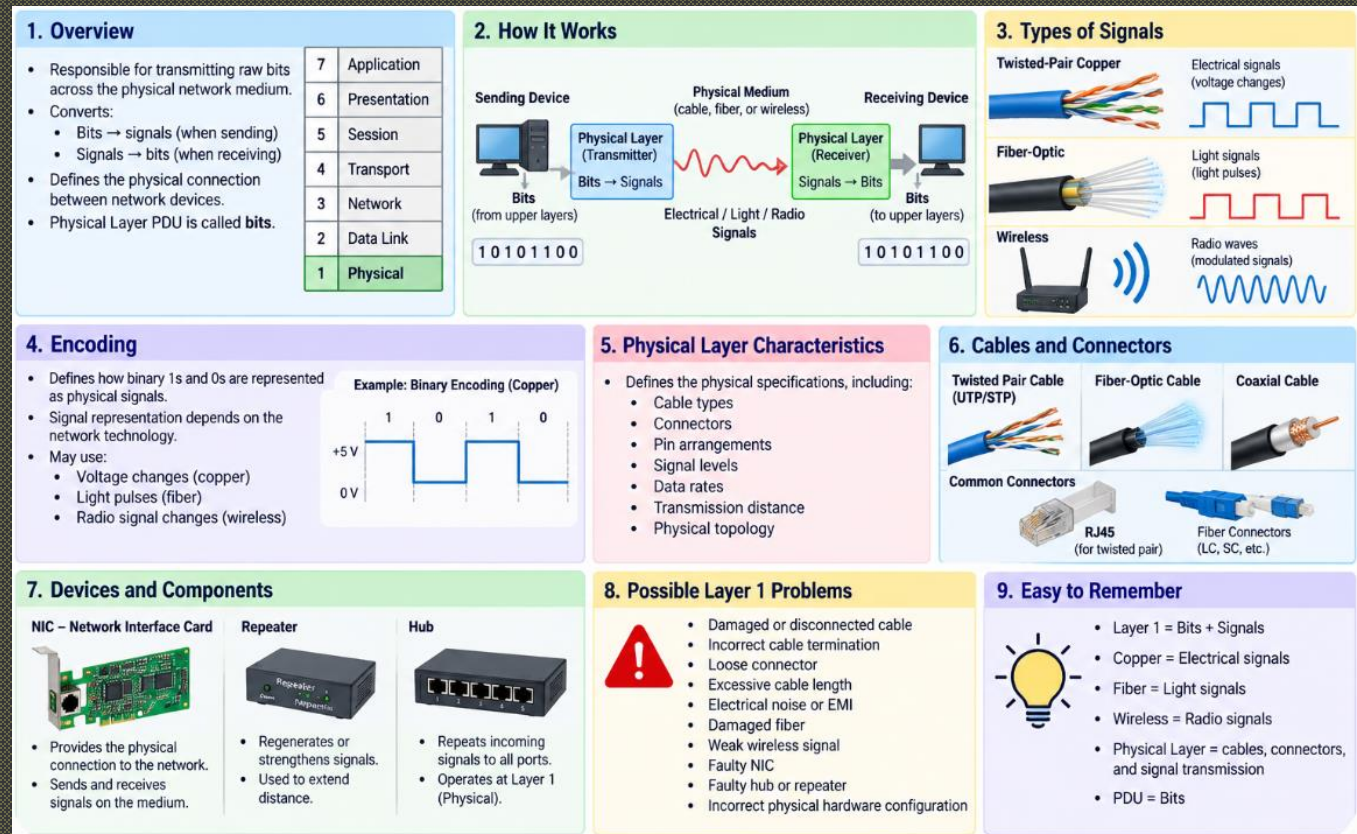
2.2 OSI REFERENCE MODEL

Physical Layer

- Responsible for transmitting raw bits across the physical network medium.
- Converts:
 - Bits → signals when sending
 - Signals → bits when receiving
- Defines the physical connection between network devices.
- Physical Layer PDU is called bits.

Types of Signals

- Twisted-pair copper cable
 - Uses electrical signals.
- Fiber-optic cable
 - Uses light signals.
- Wireless communication
 - Uses radio waves.



2.2 OSI REFERENCE MODEL

Physical Layer

Encoding

- Defines how binary 1s and 0s are represented as physical signals.
- Signal representation depends on the network technology.
- May use:

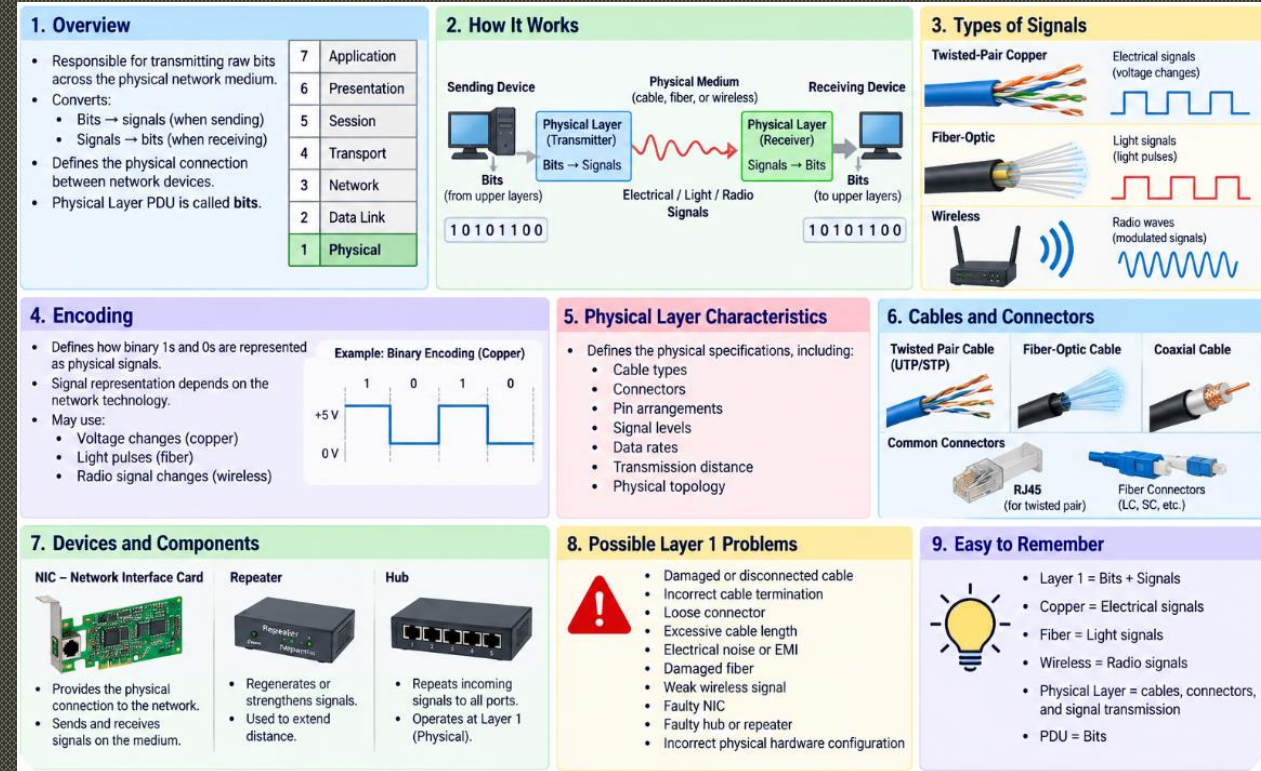
- Voltage changes
- Light pulses
- Radio signal changes

Physical Layer Characteristics

- Defines:
 - Cable types
 - Connectors
 - Pin arrangements
 - Signal levels
 - Data rates
 - Transmission distance
 - Physical topology

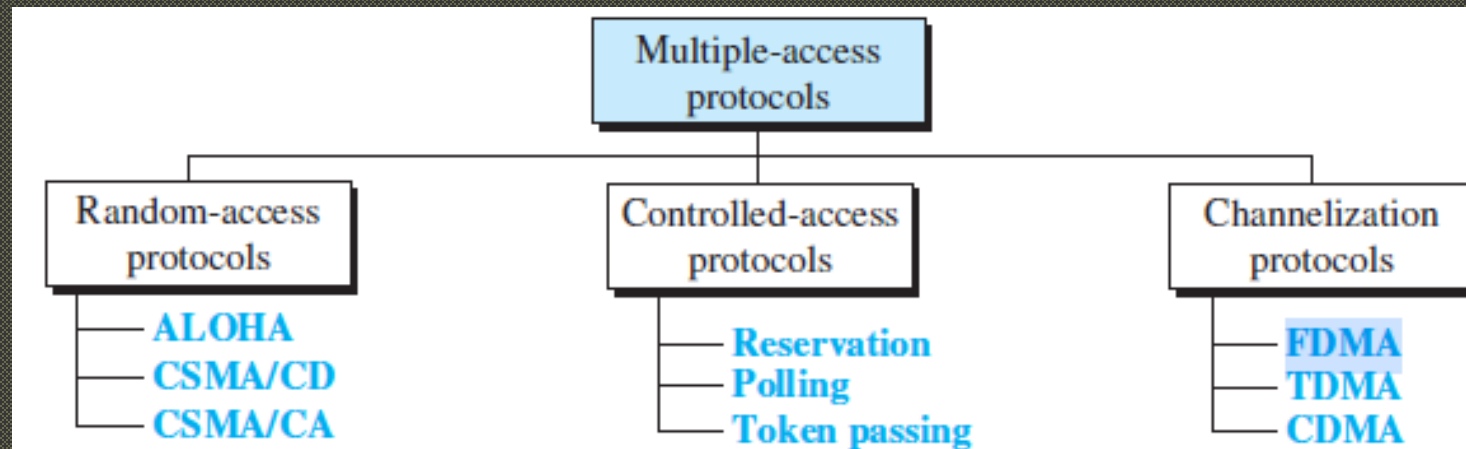
Devices and Components

- Cables
 - Twisted pair
 - Fiber optic
 - Coaxial
- Connectors
 - RJ45
 - Fiber connectors
- NIC – Network Interface Card
 - Provides the physical connection to the network.
- Repeater
 - Regenerates or strengthens signals.
- Hub
 - Repeats incoming signals to connected ports.



2.3 MEDIA ACCESS PROTOCOLS

- Multiple devices may need to share the same communication channel.
- Access protocols determine who can transmit and when.
- Three main approaches:
 - Channelization → divide the channel
 - Controlled Access → devices take turns
 - Random Access → devices compete for access
- Each method attempts to improve efficiency and reduce communication conflicts.



2.3 MEDIA ACCESS PROTOCOLS

Channelization

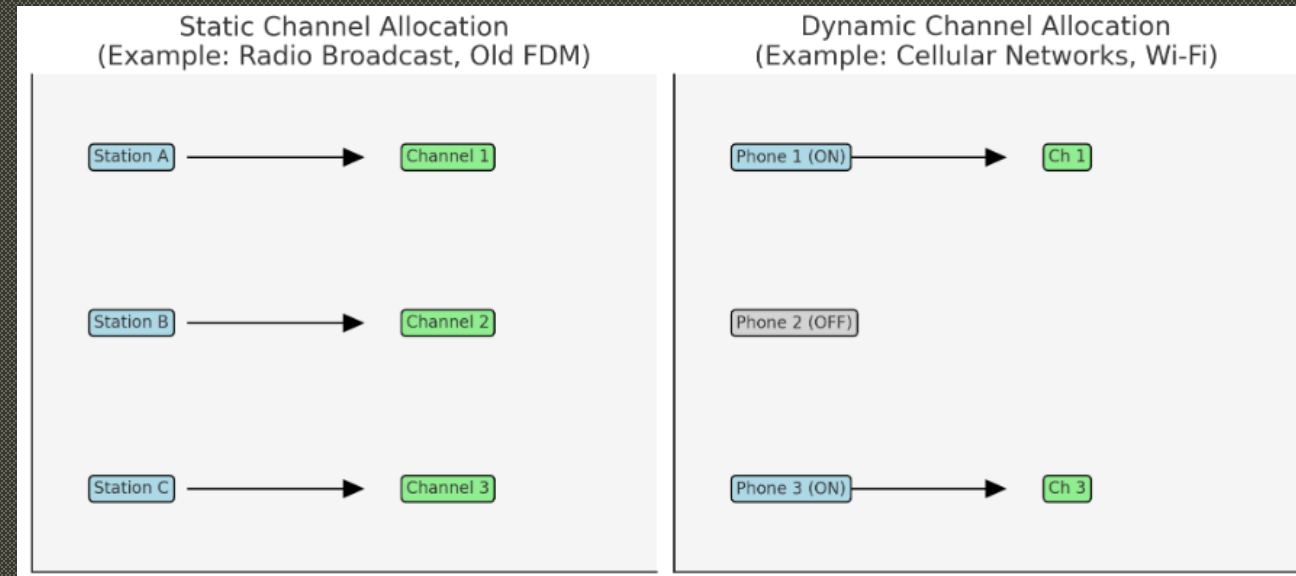
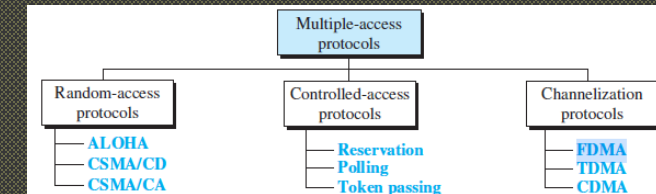
- Channelization divides a communication medium so several users can communicate.

Static Allocation

- A channel is permanently assigned to a particular user.
- Works well when:
 - Number of users is known
 - Users do not change frequently
- Example:
 - Fixed frequency assignments

Dynamic Allocation

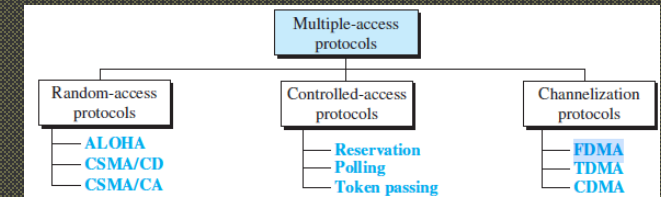
- Channels are assigned only when users need them.
- Channels can be released and reused.
- More efficient when users enter and leave the network.
- Example:
 - Cellular communication systems.



- Static → users always keep the same fixed channel.
- Right: Dynamic → channels are only assigned when a user is active (ON).

2.3 MEDIA ACCESS PROTOCOLS

Channelization - FDMA



FDMA – Frequency Division Multiple Access

- Divides the available frequency range into separate frequency channels.
- Each active user is assigned a different frequency band.
- Multiple users can transmit at the same time on different frequencies.

Advantages

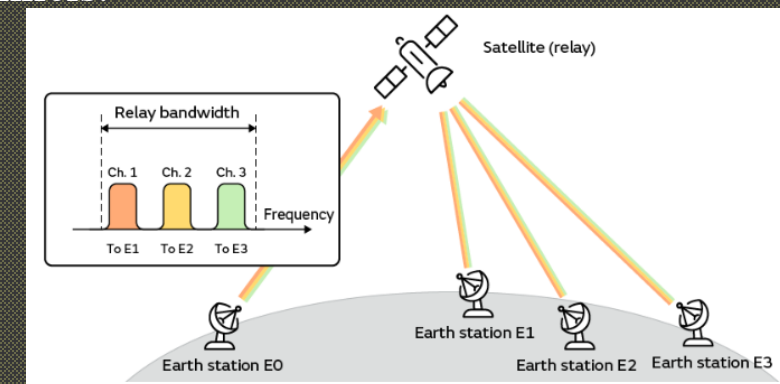
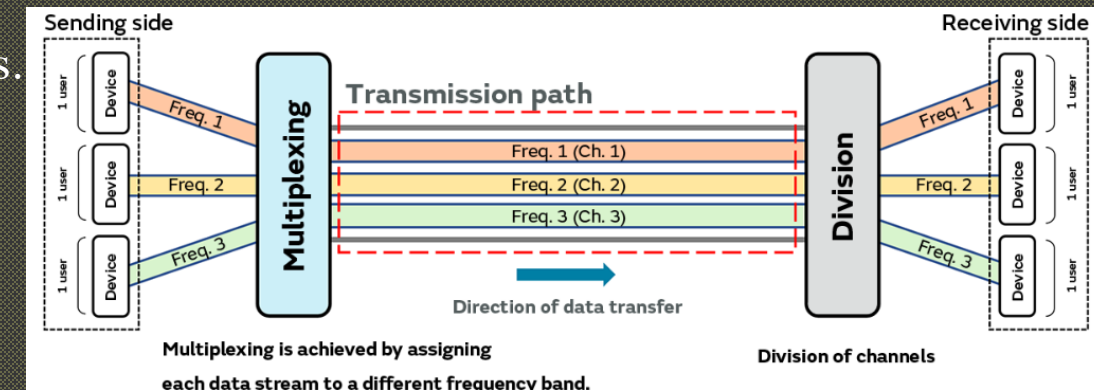
- Simple concept
- Little interference between properly separated channels
- No collisions between users assigned different channels

Disadvantages

- Assigned bandwidth can be wasted when a user is idle.
- Number of simultaneous users is limited by available frequency channels.

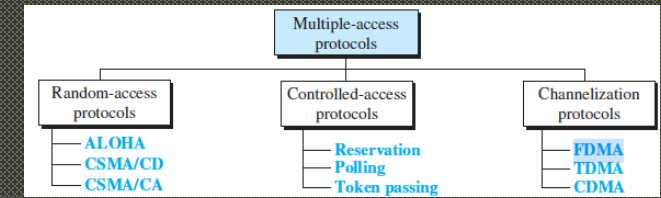
Examples

- Early analog cellular networks
- Satellite communication
- Radio systems.



2.3 MEDIA ACCESS PROTOCOLS

Channelization - TDMA



TDMA – Time Division Multiple Access

- Divides communication into repeating time slots.
- Each user transmits during its assigned time slot.
- Users share the same frequency but transmit at different times.

Advantages

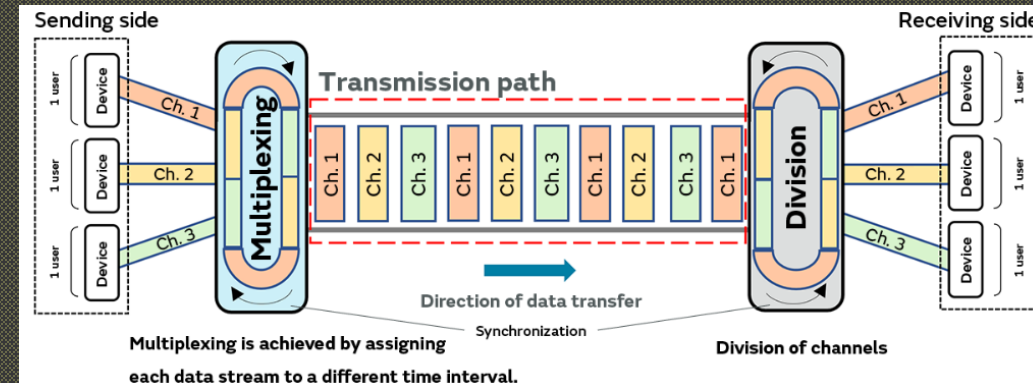
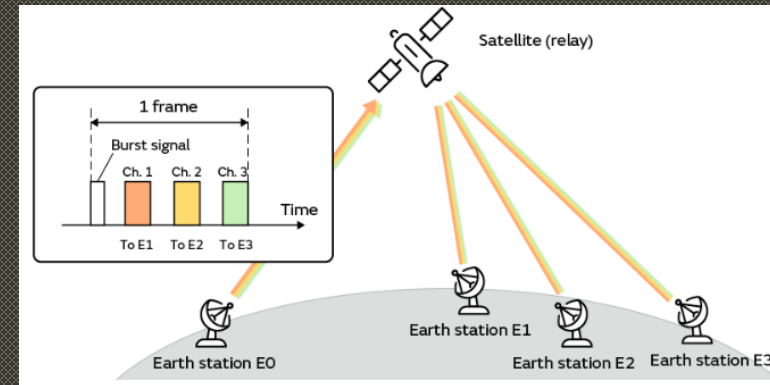
- Bandwidth can be used efficiently.
- Time slots can be reassigned to active users.

Disadvantages

- Requires accurate timing and synchronization.

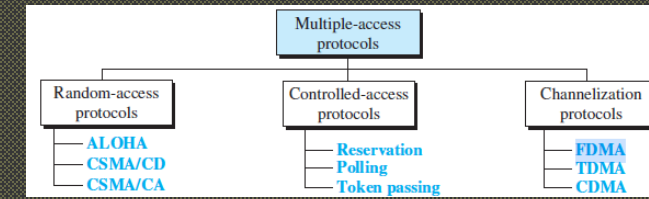
Examples

- 2G GSM cellular networks
- Some digital radio systems.
- GSM → Global Systems for Mobile Communications.



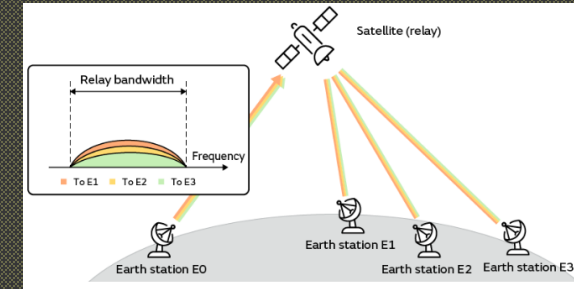
2.3 MEDIA ACCESS PROTOCOLS

Channelization - CDMA



CDMA – Code Division Multiple Access

- Multiple users can transmit:
 - At the same time
 - On the same frequency
- Each user is assigned a unique code.
- The receiver uses the code to separate one user's signal from the others.



Advantages

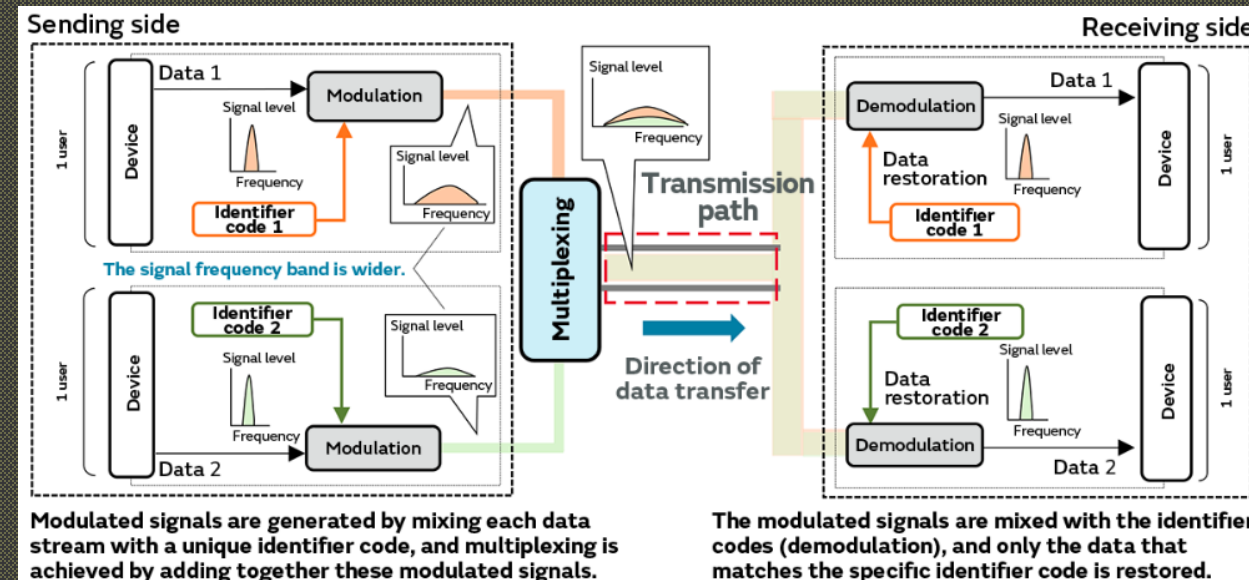
- Supports many users.
- Efficient use of available spectrum.
- Works well in mobile communication environments.

Disadvantages

- More complex than FDMA or TDMA.
- Requires sophisticated signal processing.
- Power differences between users must be carefully managed.

Examples

- 3G cellular networks
- GPS/spread-spectrum systems
- Some military communication systems.



<https://article.murata.com/en-global/article/multiplexing-and-multiple-access-1>

2.3 MEDIA ACCESS PROTOCOLS

Control Access - Polling

- Controlled access gives devices organized turns to transmit.
- Polling uses a central controller.
- The controller asks each device whether it needs to transmit.

Round-Robin Polling

- Every device receives a turn in sequence.
- Provides fair access.

Priority Polling

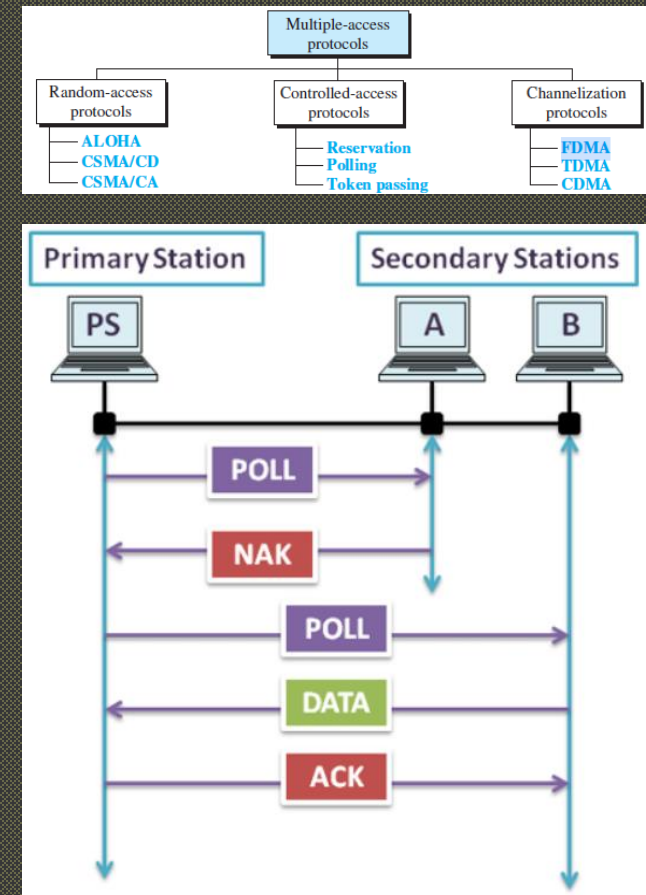
- Higher-priority devices may transmit before lower-priority devices.

Advantages

- Avoids collisions.
- Predictable access.
- Can support priority traffic.

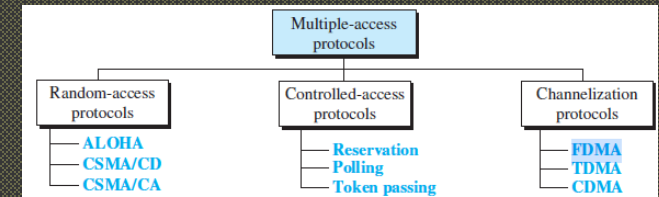
Disadvantages

- Central controller can become a bottleneck.
- Controller failure may stop communication.
- Time may be wasted polling devices that have nothing to send.



2.3 MEDIA ACCESS PROTOCOLS

Control Access - Reservation



- Devices reserve transmission opportunities in advance.
- A device first indicates that it has data to send.
- Transmission time or slots are then assigned.
- The device transmits during its reserved opportunity.

Advantages

- Avoids collisions.
- Efficient when many devices are active.
- Provides organized access.

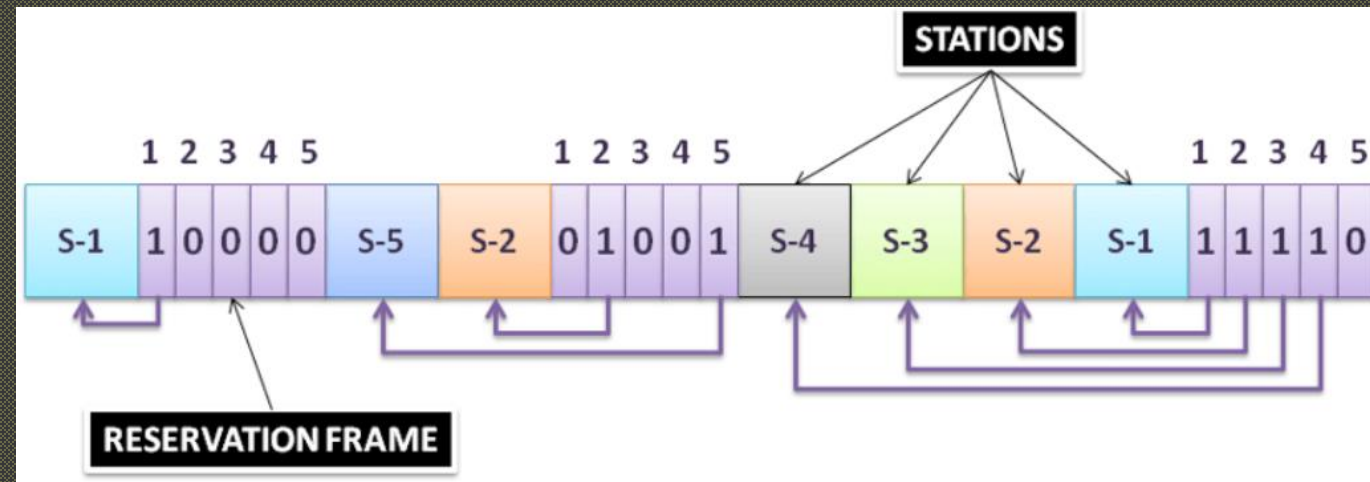
Disadvantages

- Reservation process adds overhead.
- Devices must wait for their assigned opportunity.

Examples

- Satellite networks
- Scheduled cellular uplinks
- Some DOCSIS/cable communication systems.

DOCSIS - Data Over Cable Service Interface Specification

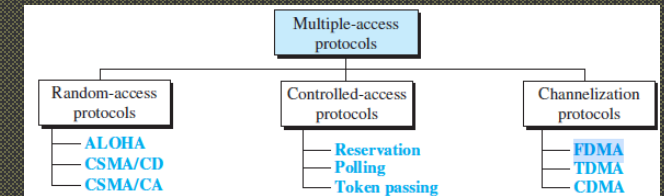


<https://www.sanfoundry.com/controlled-access-protocols-in-computer-network/>

- 5 stations share a communication channel using a mini-slot reservation system.
- Interval 1: Stations 1, 2, 3, and 4 send data in their reserved slots.
- Interval 2: Only stations 2 and 5 send data.
- Interval 3: Only station 1 sends data.
- When a station transmits in its slot, no other station can use the channel until that slot ends.

2.3 MEDIA ACCESS PROTOCOLS

Control Access - Token Passing



- A special control message called a token moves between devices.
- Only the device holding the token is allowed to transmit.
- When transmission is complete, the token is passed to another device.

Advantages

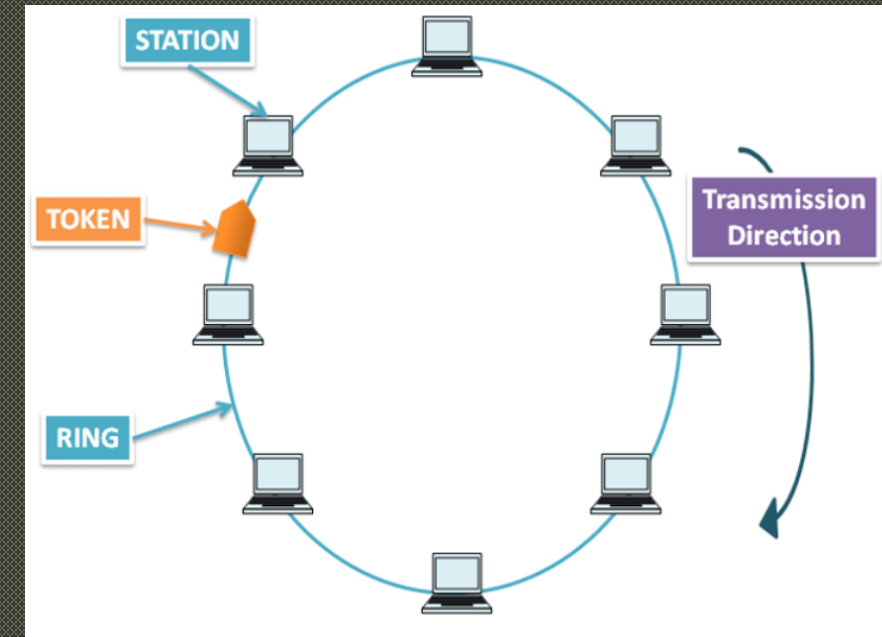
- Prevents collisions.
- Provides fair and predictable access.

Disadvantages

- Token management adds overhead.
- A lost or damaged token must be recovered.

Examples

- IBM Token Ring
- Token Bus
- Token-based industrial systems such as PROFIBUS master communication.



<https://www.sanfoundry.com/controlled-access-protocols-in-computer-network/>

2.3 MEDIA ACCESS PROTOCOLS

Random Access - ALOHA

Random Access

- Devices are not assigned fixed transmission turns.
- A device attempts to transmit when it has data.
- Two devices may transmit at the same time and cause a collision.
- Devices use random waiting periods before trying again.

ALOHA

- One of the earliest random-access protocols.
- A device sends whenever it has data.
- If acknowledgement is not received:

- Wait a random amount of time
- Retransmit

Pure ALOHA

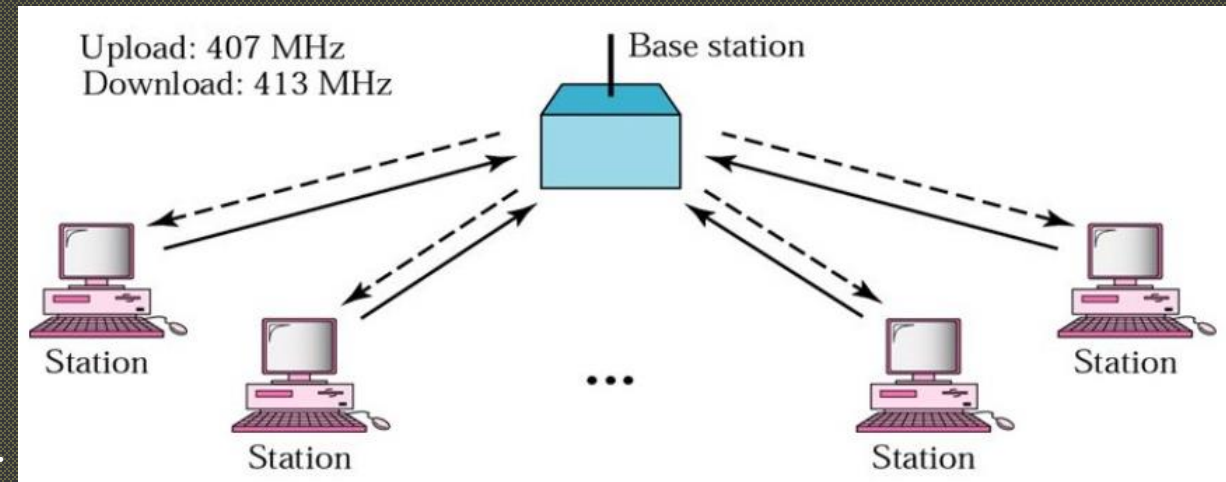
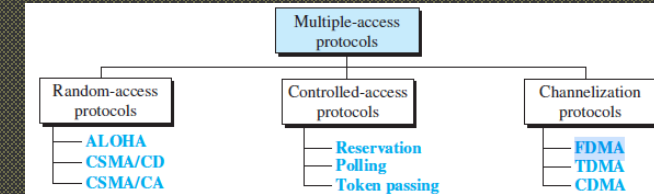
- Maximum theoretical utilization is approximately 18%.

Slotted ALOHA

- Devices transmit only at defined time boundaries.
- Maximum theoretical utilization is approximately 37%.

Importance

- ALOHA helped establish ideas later used in contention-based networking.



https://www.ques10.com/p/37482/explain-alo-ha-and-slotted-alo-ha/#google_vignette

2.3 MEDIA ACCESS PROTOCOLS

Random Access - CSMA/CD

CSMA/CD – Carrier Sense Multiple Access with Collision Detection

- Used in traditional shared, half-duplex Ethernet.

How It Works

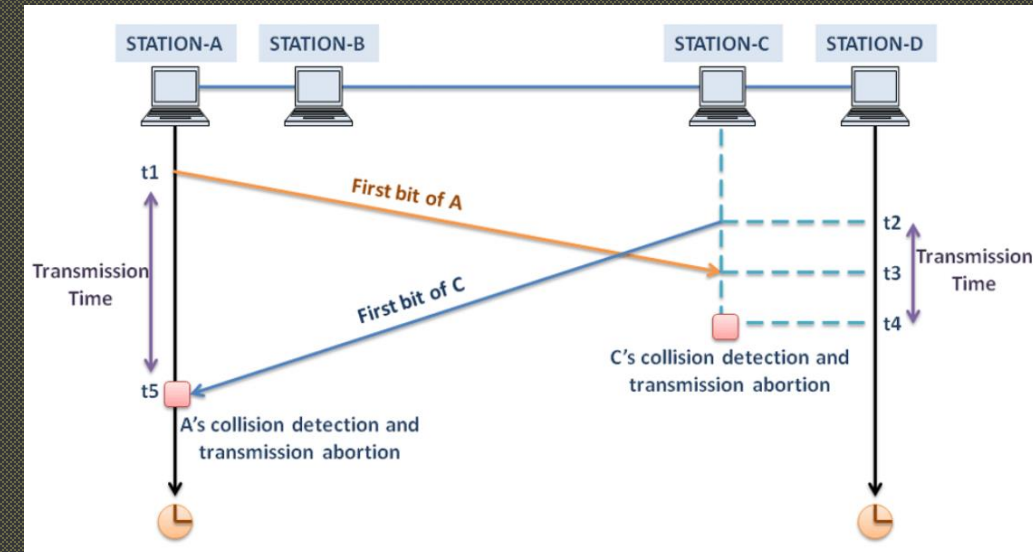
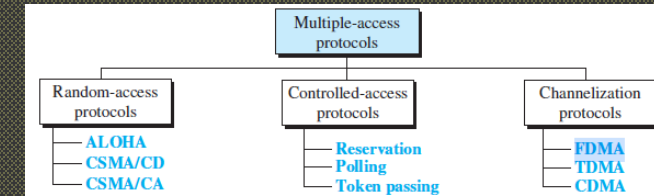
- Carrier Sense
 - Device listens to determine whether the medium is busy.
- If the medium is idle:
 - Device begins transmitting.
- Collision Detection
 - Device monitors for a collision while transmitting.
- If a collision occurs:
 - Transmission stops.
 - Devices wait a random amount of time.
 - They try again later.
- Uses binary exponential backoff to reduce repeated collisions.

Advantages

- Efficient for traditional shared Ethernet.

Modern Ethernet

- Switched full-duplex Ethernet eliminates normal Ethernet collisions.
- Therefore CSMA/CD is generally not needed in modern switched networks.



<https://www.sanfoundry.com/computer-network-random-access-protocols-aloha-csma-csma-ca-csma-cd/>

- Station-A begins transmitting at time t1.
- Station-C, unaware of A's transmission, starts transmitting at t2.
- A collision occurs when their signals meet.
- Station-C detects the collision first at t4 and stops.
- Station-A detects the collision later at t5 and stops.
- Transmission durations: A for (t5-t1), C for (t4-t2).

2.3 MEDIA ACCESS PROTOCOLS

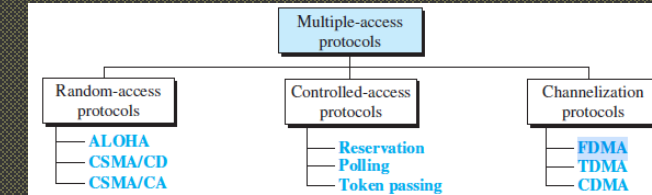
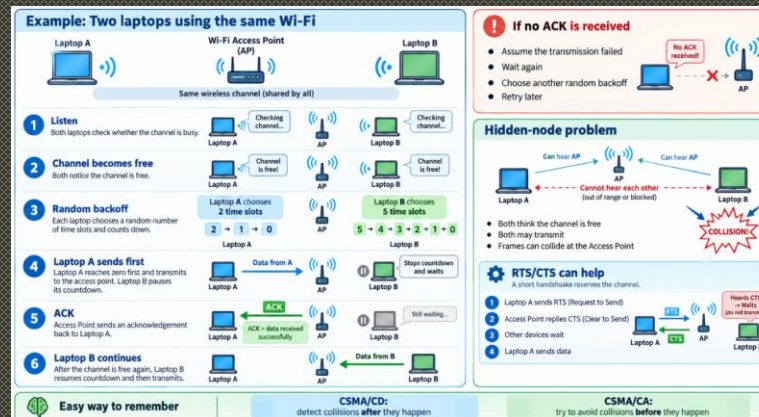
Random Access - CSMA/CA

CSMA/CA – Carrier Sense Multiple Access with Collision Avoidance

- Used primarily in Wi-Fi – IEEE 802.11.
- Wireless devices cannot reliably detect collisions while transmitting.
- Therefore Wi-Fi tries to avoid collisions before transmission.

How It Works

- Device listens to determine whether the wireless channel is busy.
- If the channel is available:
 - The device waits according to the access rules.
 - Selects a random backoff time.
 - Transmits when its backoff timer reaches zero.
- Receiver sends an ACK when data is received successfully.
- If no ACK is received:
 - Sender assumes transmission failed.
 - It waits and retries.



RTS/CTS

- Wi-Fi can optionally use:
 - RTS – Request to Send
 - CTS – Clear to Send
- RTS/CTS can help reduce the hidden-node problem.
- Other stations hearing RTS/CTS wait before transmitting.

Advantages

- Suitable for wireless networks.
- Reduces the probability of collisions.
- Helps manage hidden wireless stations.

Disadvantages

- Backoff, acknowledgements, and optional RTS/CTS add overhead.

Modern Use

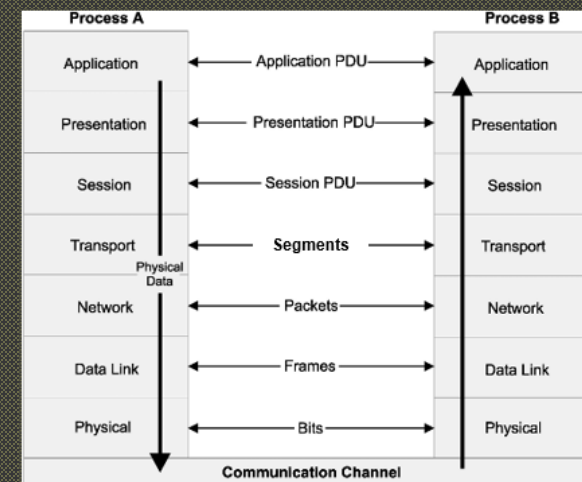
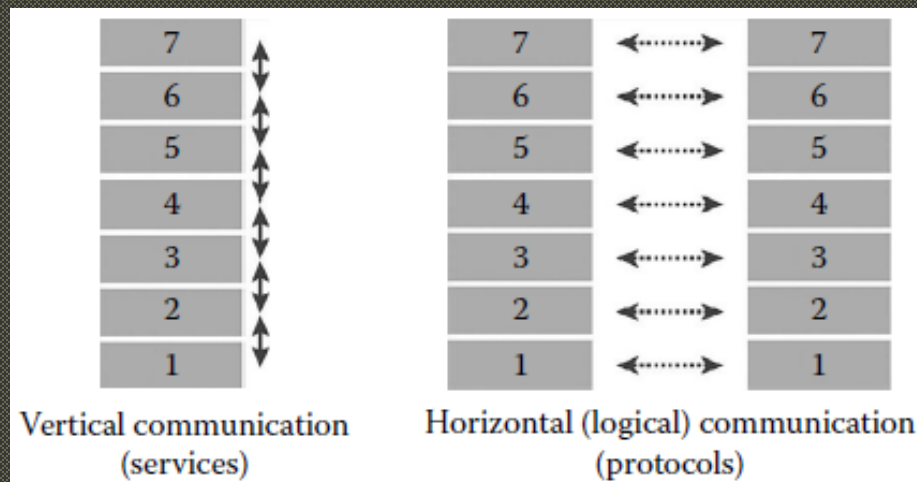
- Fundamental access method used by Wi-Fi networks.

2.4 ENCAPSULATION PROCESS

Interfaces and Peer Communication

- Vertical communication
 - Data moves between adjacent layers inside the same device.
 - Each layer uses services provided by the layer below it.
 - Sending data moves down the stack.
 - Receiving data moves up the stack.
- Horizontal / peer communication
 - The same layer on two different devices logically communicates using the same protocol rules.
 - Example:
 - Transport Layer ↔ Transport Layer
 - Network Layer ↔ Network Layer

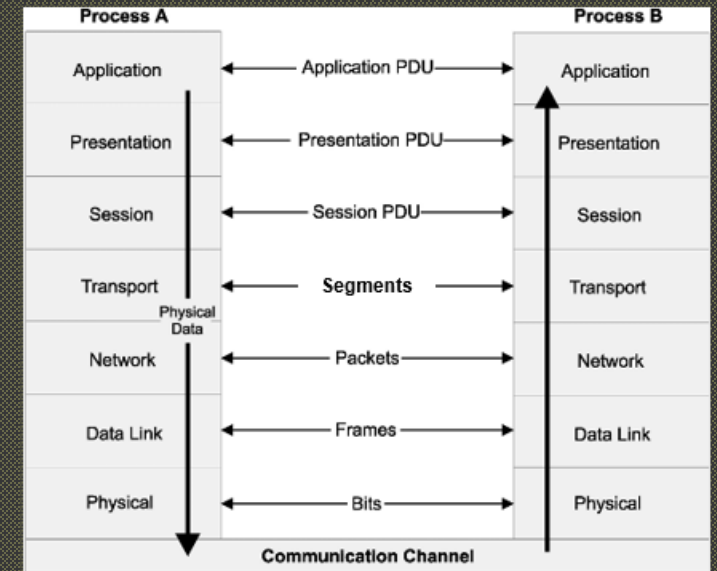
Peer communication is logical; the actual data still travels through the complete stack and network medium.



2.4 ENCAPSULATION PROCESS

OSI Layers in the Encapsulation Process

- Each OSI layer performs a different function as data travels through the stack.
- The important idea for encapsulation is:
 - Upper layers create the application data.
 - Transport prepares it for application-to-application delivery.
 - Network adds logical addressing.
 - Data Link prepares it for local network delivery.
 - Physical converts it into signals for transmission.
- The diagram shows that each peer layer interprets the information added by its corresponding layer on the sending device.



2.4 ENCAPSULATION PROCESS

Encapsulation & De-encapsulation

Encapsulation – Sender

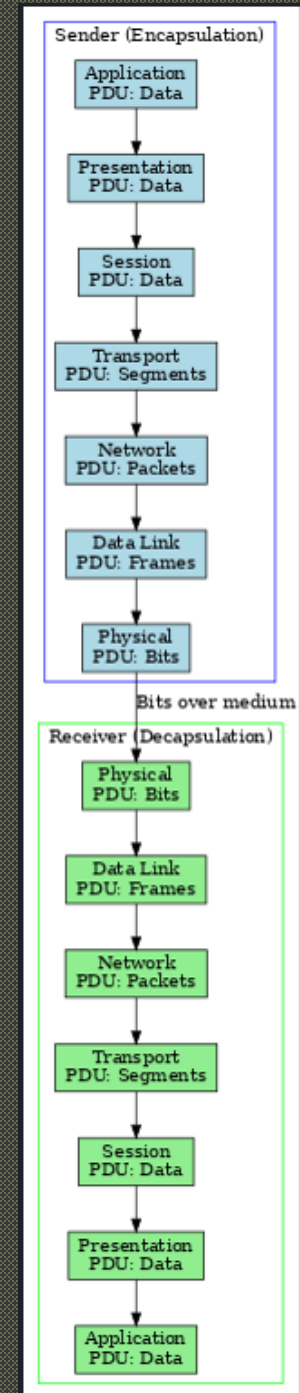
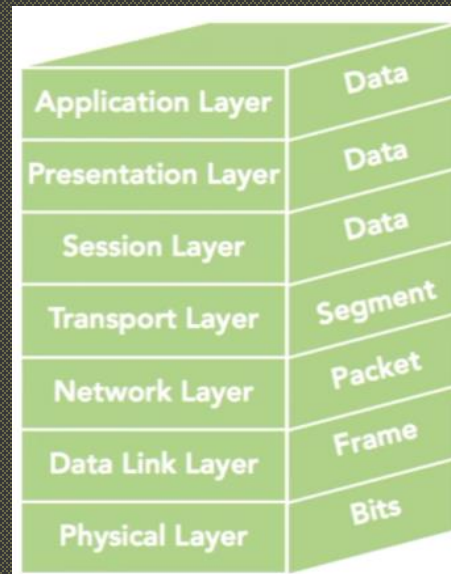
- Data travels down the protocol stack.
- Layers add control information needed for communication.
- This process is called encapsulation.

De-encapsulation – Receiver

- Data travels up the protocol stack.
- Each layer processes and removes the information intended for that layer.
- The original application data is eventually delivered to the receiving application.

PDU Names

- Layers 7–5 → Data
- Layer 4 → Segment / UDP datagram
- Layer 3 → Packet
- Layer 2 → Frame
- Layer 1 → Bits



2.4 ENCAPSULATION PROCESS

Headers, Trailers, and Data Splitting

- Encapsulation adds control information around the original data.

Header information may include:

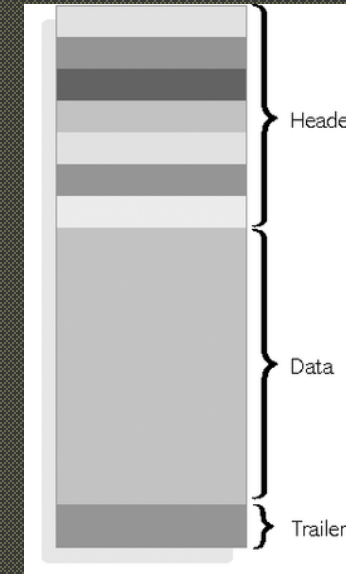
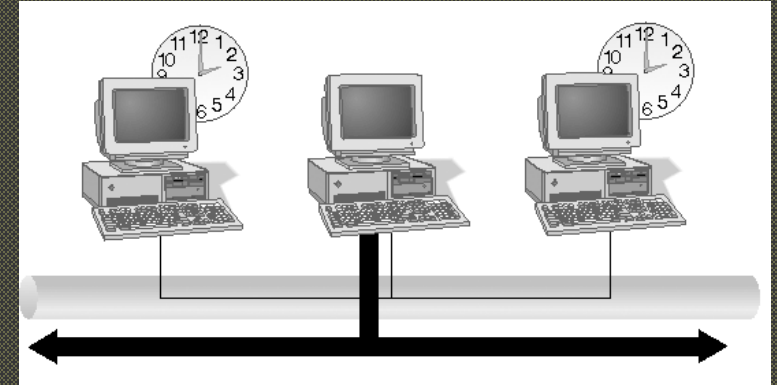
- Addresses
- Protocol identification
- Port numbers
- Sequence/control information

Trailer information may include:

- Error-detection information
- Example:
 - Ethernet FCS / CRC

Why data is divided into smaller units

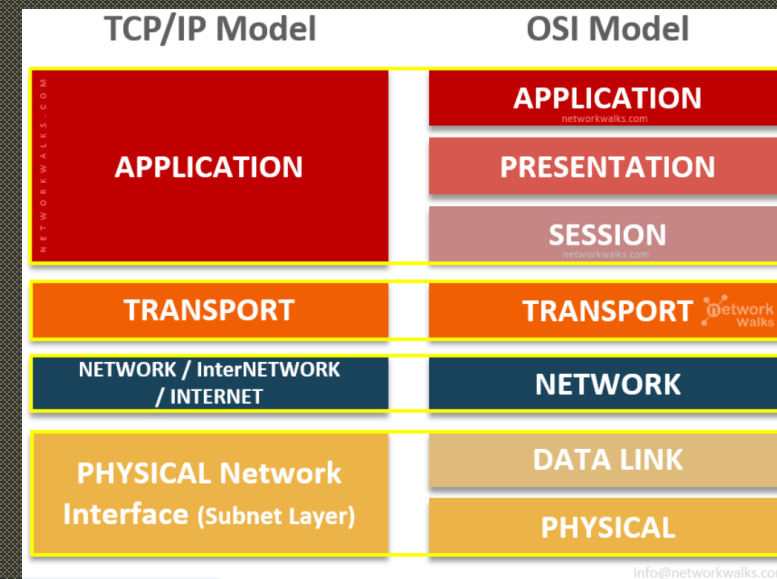
- Makes transmission easier to manage.
- Allows different network communications to share the medium.
- If part of a transmission is lost or corrupted, protocols can recover the affected data rather than necessarily repeating an entire large transfer.



2.4 ENCAPSULATION PROCESS

Encapsulation in Modern Networks

- The OSI model remains useful as a conceptual framework for understanding communication.
- Actual Internet and industrial Ethernet networks normally use the TCP/IP protocol suite.
- Encapsulation still occurs even though TCP/IP groups the OSI layers differently.
- The concept is especially useful for troubleshooting:
 - Application problem
 - Transport problem
 - IP/routing problem
 - Ethernet/physical problem.



2.4 ENCAPSULATION PROCESS

Sender and Receiver Process

Sending Device

- Application creates the data.
- Data moves downward through the stack.
- Each relevant layer adds information needed by its peer layer.
- The final frame is converted into signals/bits for transmission.

Receiving Device

- Signals are received.
- Data moves upward through the stack.
- Each layer processes its corresponding control information.
- The original data is delivered to the application.



Easy sequence

Sender:

Data → Segment → Packet → Frame → Bits

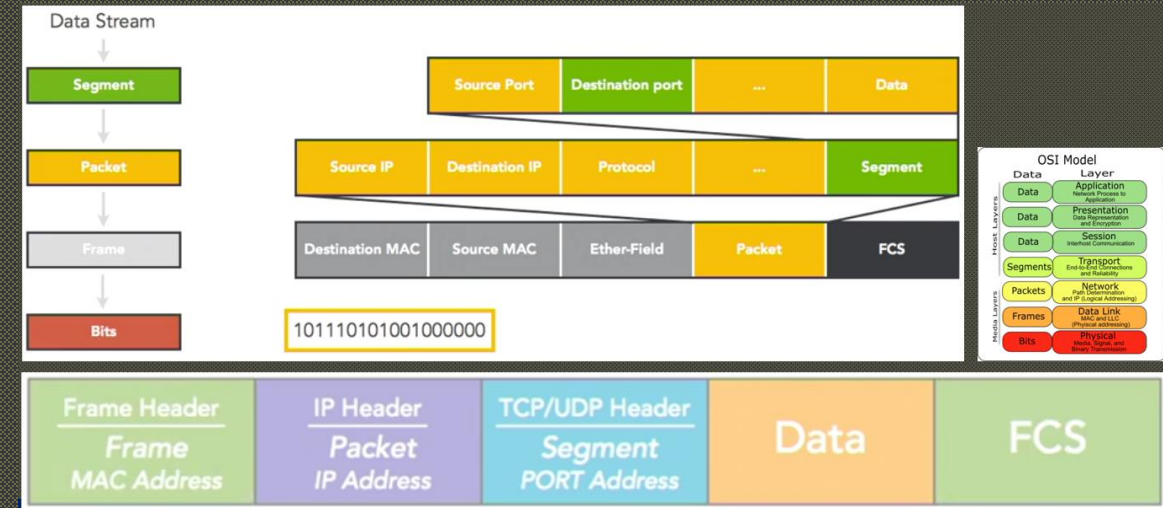
Receiver:

Bits → Frame → Packet → Segment → Data

2.4 ENCAPSULATION PROCESS

Transport Layer Encapsulation

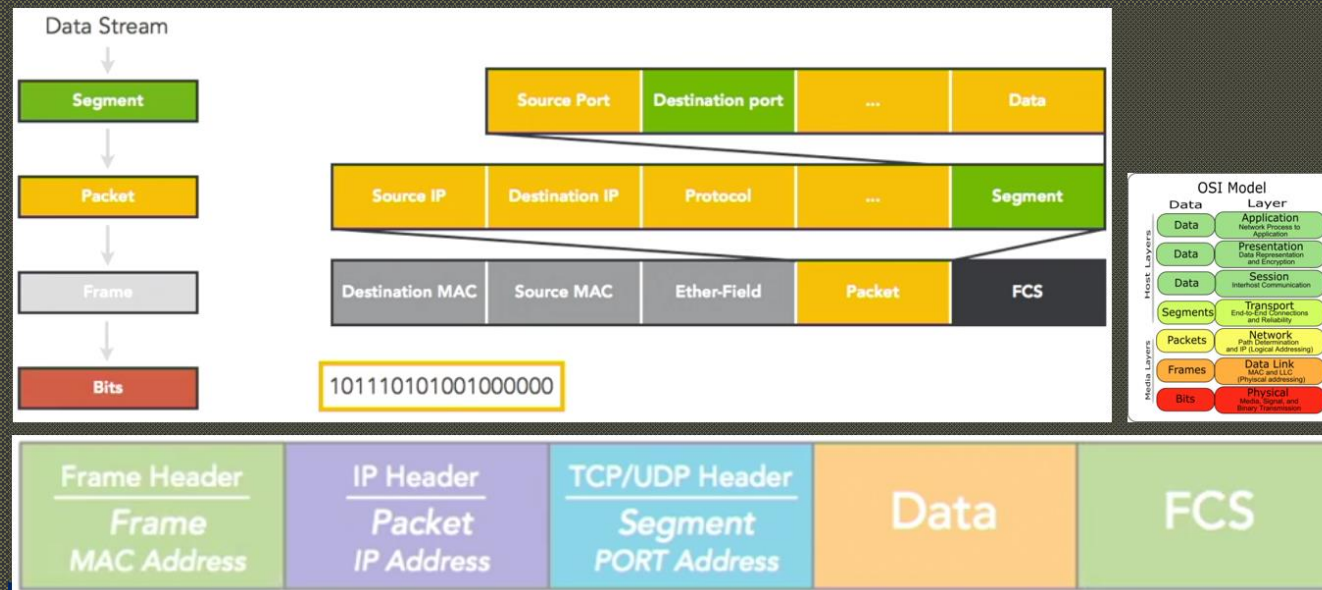
- The Transport Layer receives application data.
- It adds information such as:
 - Source port
 - Destination port
- With TCP, additional information supports functions such as:
 - Sequencing
 - Acknowledgement
 - Flow control
- With UDP, the header is simpler and has less overhead.
- QUIC is a modern transport technology that operates over UDP and is used by HTTP/3.
- The resulting Transport Layer unit is passed down to the Network Layer.



2.4 ENCAPSULATION PROCESS

Network Layer Encapsulation

- The Network Layer receives the Transport Layer data.
- It adds an IP header.
- Important information includes:
 - Source IP address
 - Destination IP address
 - Other information required for packet forwarding
- The result is an IP packet.
- Routers use Layer 3 information to forward the packet toward the destination.
- IPv4 and IPv6 are the main IP versions used in networks.



2.4 ENCAPSULATION PROCESS

Data Link and Physical Encapsulation

Data Link Layer

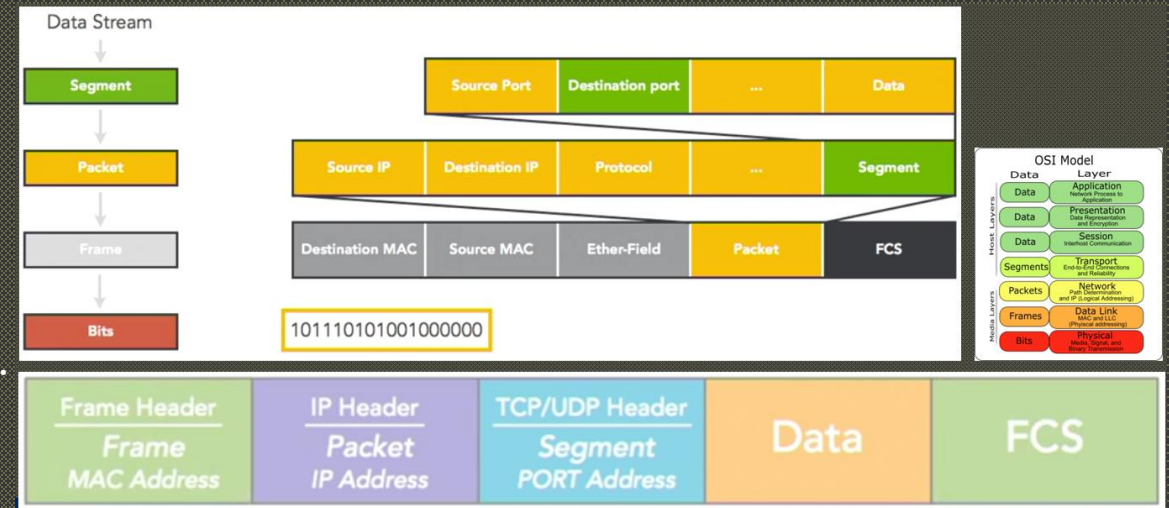
- Receives the IP packet.
- Adds an Ethernet frame header containing information such as:
 - Source MAC address
 - Destination MAC address
 - EtherType
- Adds an FCS trailer for error detection.
- The result is an Ethernet frame.

Physical Layer

- Converts the frame into a physical signal representation.
- Sends the information as:
 - Electrical signals over copper
 - Light signals over fiber
 - Radio signals over wireless media.

Encapsulation – Simple Overall Summary

- Application creates data.
 - Transport adds port information → Segment / Datagram.
 - Network adds IP addresses → Packet.
 - Data Link adds MAC addresses and FCS → Frame.
 - Physical converts the frame into bits/signals.
- The receiving computer performs the process in reverse.



Updates

- IPv6 Adoption: Now ~48% of global traffic; includes mandatory IPsec and 128-bit addressing.
- Ethernet Evolution:
 - 800 GbE common in data centers.
 - Work underway on 1.6 Terabit Ethernet.
 - Updated PoE standards powering IoT devices efficiently.
- Transport Layer:
 - QUIC widely used in HTTP/3.
 - Improves speed and security with encryption + multiplexing.

2.5 NETWORK INTERFACE CARD

- NIC = Network Interface Card
- Hardware that connects a computer or other device to a network.
- Provides the interface between:
 - The device
 - The network medium
- The network medium may be:
 - Copper Ethernet
 - Fiber
 - Wireless

NIC Installation

- Many computers have a NIC built into the motherboard.
- Additional NICs may be installed when:
 - The built-in NIC fails.
 - More network ports are required.
 - Redundancy is required.
 - Higher performance is required.
 - A different network technology is needed.

Modern NIC Features

- May support:
 - 1 Gb/s Ethernet
 - 2.5 Gb/s
 - 5 Gb/s
 - 10 Gb/s and higher
- Wireless NICs may support modern Wi-Fi standards.
- Some NICs support:
 - Virtualization
 - Energy-saving features
 - Remote startup and management features such as Wake-on-LAN or PXE.



<https://ascentoptics.com/blog/nic-network/>

2.5 NETWORK INTERFACE CARD

What a NIC Does?

- A NIC works together with a device driver.
- The driver allows the operating system to communicate with the NIC.

Connection to the Network Medium

- Copper → electrical signals
- Fiber → light signals
- Wireless → radio signals



<https://ascentoptics.com/blog/nic-network/>

Receiving Data

- Receives physical signals from the network.
- Converts signals into bits.
- Reconstructs Ethernet frames.
- Checks:
 - Frame integrity
 - Destination MAC address
- Passes valid data to the operating system and upper-layer protocols.

Sending Data

- Receives data from the operating system.
- Creates the required Ethernet frame.
- Adds information such as:
 - Source MAC address
 - Destination MAC address
 - Error-detection information
- Converts the frame into signals for transmission

2.5 NETWORK INTERFACE CARD

MAC address

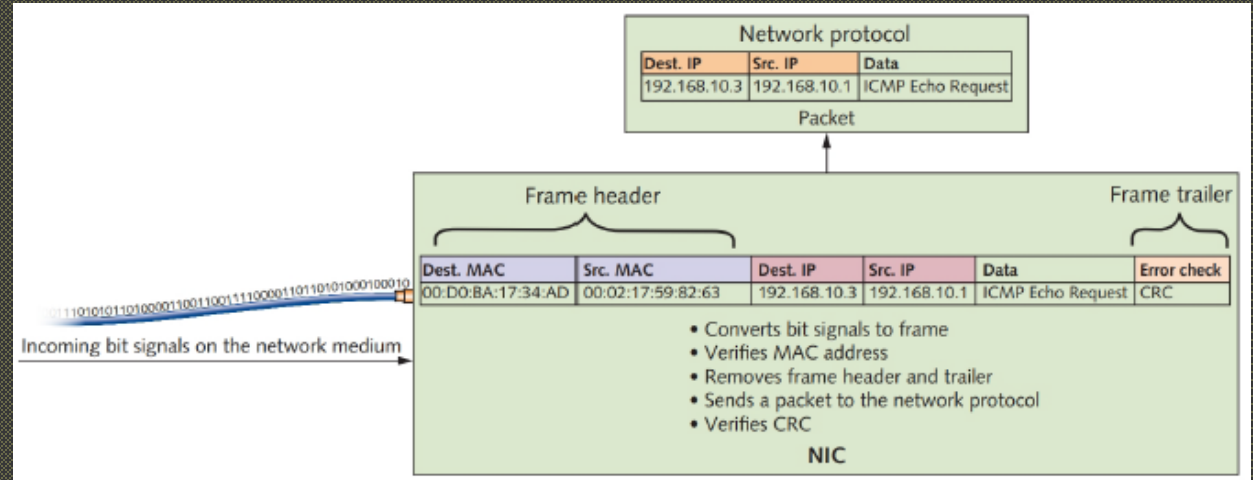
- Ethernet NICs use a MAC address for Layer 2 communication.
- A standard MAC address is normally:
 - 48 bits
 - 6 bytes
 - Written in hexadecimal
- Example:
 - 04:40:31:5B:1A:C4

MAC Address Structure

- A globally assigned MAC address commonly contains:
 - An IEEE-assigned organizational portion associated with the vendor.
 - A portion assigned by the manufacturer to the interface.

Purpose

- MAC addresses help switches and network interfaces determine where Ethernet frames should be delivered.



Modern Systems

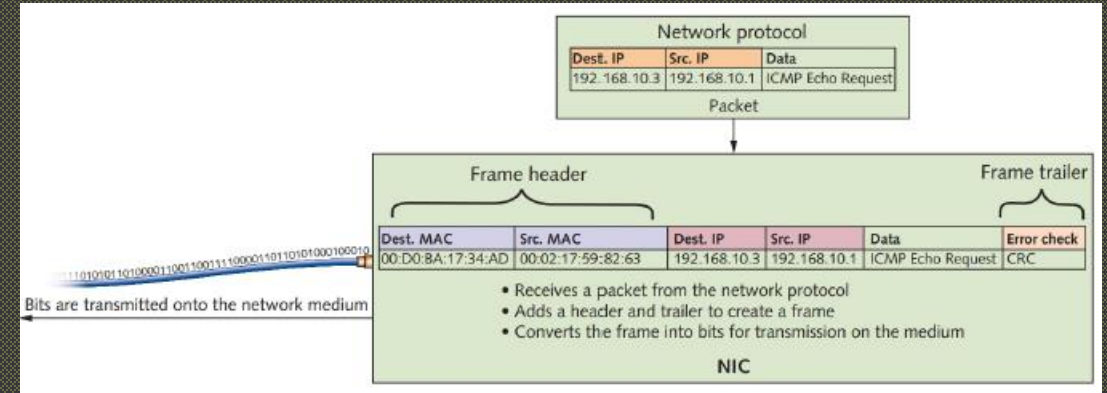
- MAC addresses do not always have to remain fixed.
- Systems may use:
 - Locally administered MAC addresses
 - Virtual MAC addresses
 - Software-assigned addresses
- Virtualization can allow one physical NIC to support multiple logical network interfaces.
- Update: With virtualization and cloud computing, some NICs support virtual MAC addresses, allowing one physical NIC to host multiple virtual machines.

2.5 NETWORK INTERFACE CARD

Frame Acceptance and NIC Status

Frames Accepted by a NIC

- Unicast
 - Sent to one specific MAC address.
 - The NIC accepts frames addressed to it.
- Broadcast
 - Sent to all devices on the local broadcast domain.
 - Ethernet broadcast address:
 - FF:FF:FF:FF:FF:FF
- Multicast
 - Sent to a selected group of devices.
- Promiscuous mode
 - NIC can receive frames not specifically addressed to it.
 - Useful for:
 - Packet capture
 - Network troubleshooting
 - Tools such as Wireshark.

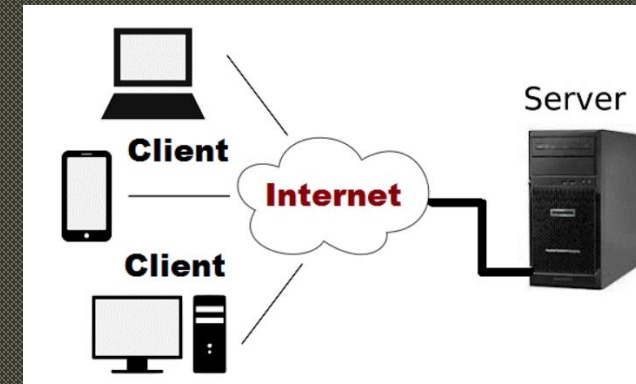


NIC Status Indicators

- NICs commonly provide LEDs for troubleshooting.
- Link indicator
 - Shows whether a physical Ethernet link has been established.
- Activity indicator
 - Flashes when data is being transmitted or received.
- Speed indicator
 - Some NICs indicate the negotiated Ethernet speed.

2.6 Clients and Servers

- A client is a device or software application that requests data, resources, or services from a server.
- The term *client* can describe three things:
 1. Client Operating System
 - Designed mainly for end users to:
 - Run applications
 - Access network resources
 - Connect to servers
 - Examples:
 - Windows 10/11, macOS, Ubuntu
 2. Client Computer
 - A physical device used by a user to access network services.
 - Examples:
 - Desktop computer, Laptop, Smartphone, Industrial workstation
 3. Client Software
 - Software that sends requests to a server.
 - Examples:
 - Web browser, Email application, File-sharing client, or Industrial engineering software connecting to a PLC or server



<https://teachcomputerscience.com/client-server-architecture/>

2.6 Clients and Servers

- A server is a device or software application that provides resources or services to clients.
- Being a server is mainly determined by the role it performs, not simply by the type of computer.

1. Server Operating System

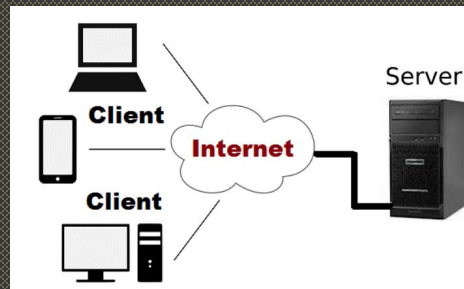
- Designed to:
 - Share resources
 - Support multiple users
 - Run network services
 - Handle client requests
- Examples:
 - Windows Server, Red Hat Enterprise Linux, Ubuntu Server

2. Server Computer

- A computer whose main purpose is to host services or resources.
- Commonly found in:
 - Server rooms
 - Data centers
 - Cloud environments
- Examples of cloud platforms:
 - AWS, Microsoft Azure, Google Cloud

3. Server Software

- The software that actually provides the service.
- Examples:
 - Web server
 - Apache, Nginx, Microsoft IIS
 - Email server
 - Microsoft Exchange, Postfix
 - File servers
 - Database servers
 - Printer-sharing services
- In Windows, server programs running in the background are often called services.
- In Linux/UNIX, background services are often called daemons.



<https://teachcomputerscience.com/client-server-architecture/>

2.6 Clients and Servers

Client/Server Communication

- The client sends a request.
- The server receives and processes the request.
- The server sends a response back to the client.

Simple example

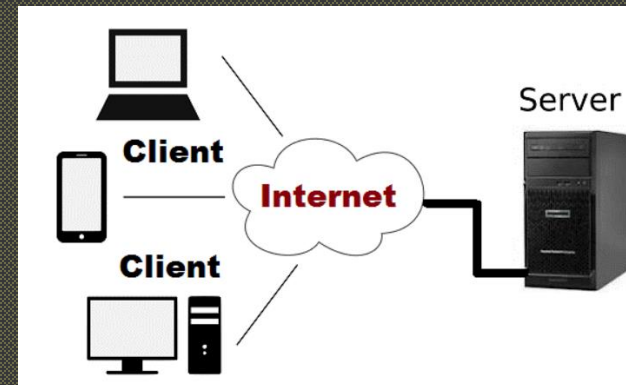
- Web browser → requests a webpage → Client
- Web server → provides the webpage → Server

Industrial example

- An HMI or engineering workstation may act as a client.
- A PLC, OPC server, database server, or web server may provide information or services.
- Example:
 - HMI requests PLC data.
 - PLC provides the requested process information.

A Device Can Be Both Client and Server

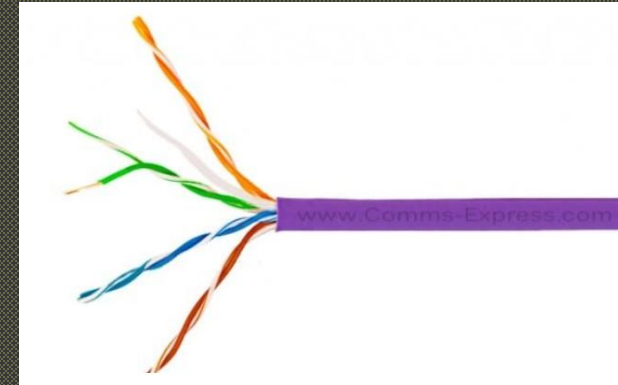
- A computer does not always have to be only a client or only a server.
- The same device can perform both roles, depending on the communication taking place.
- For example:
 - A computer may act as a client when downloading data from another server.
 - The same computer may act as a server when sharing files with another device.



2.7 NETWORK MEDIA

Unshielded Twisted Pair (UTP)

- Common copper cable used in Ethernet LANs.
- Contains four twisted pairs of copper wires.
- Commonly connects devices to a central Ethernet switch.
- Typical maximum Ethernet distance:
 - About 100 m (328 ft).
- Common cable categories include:
 - Cat 5e
 - Cat 6
 - Cat 6A
 - Cat 7
 - Cat 8
- Higher categories generally support higher speeds and improved performance.
- Main disadvantage:
 - Copper is more affected by electrical noise and EMI than fiber.
- Copper Ethernet is commonly used for short connections between equipment and switches.



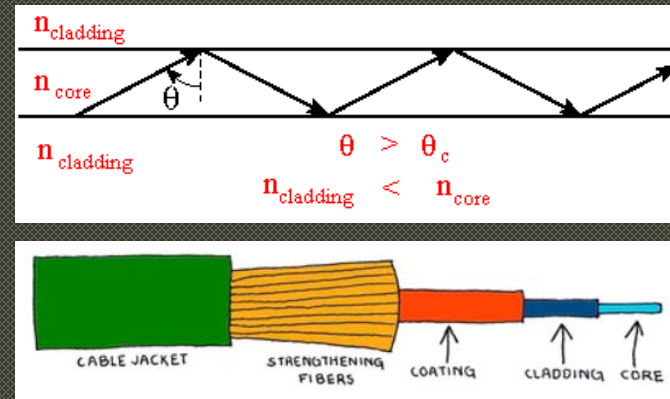
<https://www.comms-express.com/infozone/article/cat5e-wiring/>

2.7 NETWORK MEDIA

Fiber-Optic and Coaxial Cable

Fiber-Optic Cable

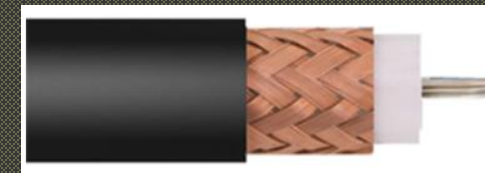
- Uses strands of glass or plastic to carry light signals.
- Supports:
 - High data rates
 - Long transmission distances
- Resistant to:
 - Electrical noise
 - Electromagnetic interference
- Commonly used for:
 - Industrial backbones
 - Data centers
 - WANs
 - Long-distance links
- Especially useful in electrically noisy industrial environments.



<https://www.truecable.com/blogs/cable-academy/basic-components-of-a-fiber-optic-cable?>

Coaxial Cable

- Contains:
 - Center conductor
 - Insulation
 - Metallic shield
 - Outer jacket
- Used in older Ethernet networks.
- Still widely used for:
 - Cable television
 - Cable Internet
 - DOCSIS systems
- Less common than twisted-pair or fiber in modern industrial Ethernet LANs.



- Central Copper Wire
- Plastic Insulator
- Mesh Conductive Layer
- Wire Sheath

<https://www.ic-components.com/blog/overview-of-coaxial-cable-basics.jsp>

2.7 NETWORK MEDIA

Baseband vs. Broadband Signaling

Baseband

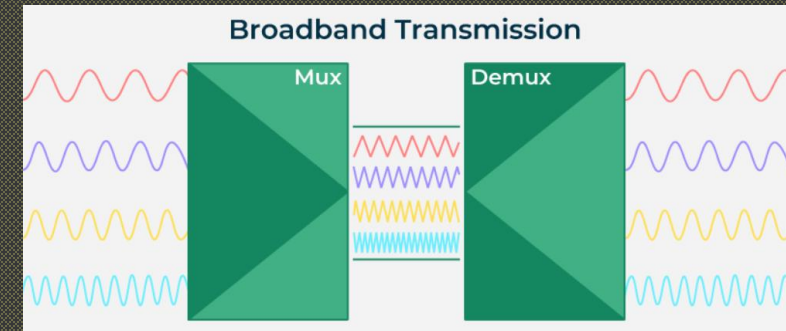
- Uses the communication medium for one digital communication system.
- Commonly associated with Ethernet LANs.
- The available medium is used directly for data communication.

Broadband

- Divides the available frequency range into multiple channels.
- Allows multiple services or signals to share the same medium.
- Common examples:
 - Cable television
 - Cable Internet

To Remember

- Baseband = one communication system uses the medium
- Broadband = multiple frequency channels share the medium



<https://www.geeksforgeeks.org/digital-logic/difference-between-broadband-and-baseband-transmission/>

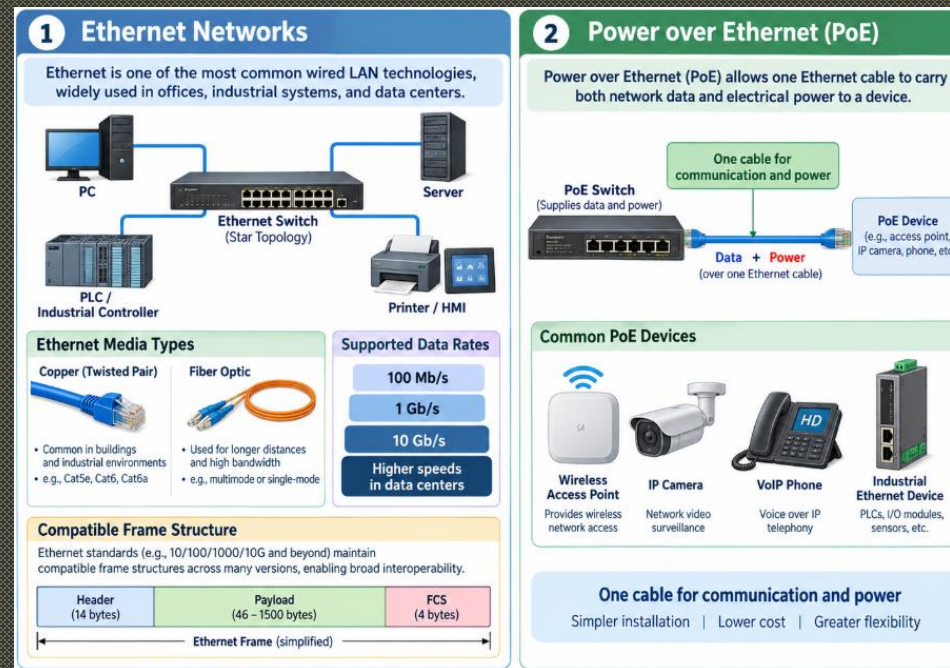
2.7 NETWORK MEDIA

Ethernet

- Ethernet is one of the most common wired LAN technologies.
- Modern Ethernet commonly uses:
 - Switches
 - Star topology
- Can operate over:
 - Copper
 - Fiber
- Supports many data rates, including:
 - 100 Mb/s
 - 1 Gb/s
 - 10 Gb/s
 - Higher speeds in data centers
- Ethernet standards maintain compatible frame structures across many versions.

Power over Ethernet (PoE)

- Allows one Ethernet cable to carry:
 - Network data
 - Electrical power
- Common PoE devices include:
 - Wireless access points
 - IP cameras
 - VoIP phones
 - Some industrial Ethernet devices.



2.7 NETWORK MEDIA

Wireless LAN (WLAN)

Wireless LAN (WLAN)

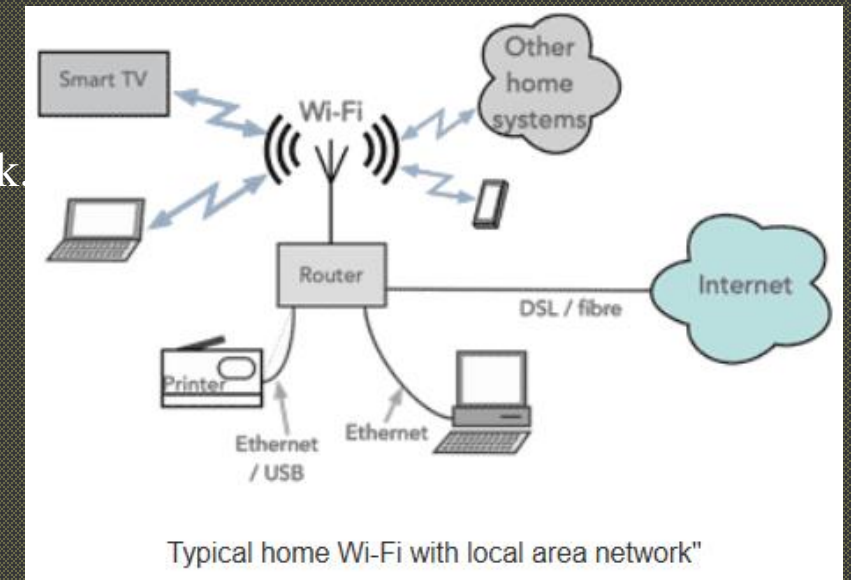
- WLAN connects devices using radio waves instead of cables.
- Wi-Fi is based on the IEEE 802.11 family of standards.
- A wireless device uses a wireless NIC.
- The NIC converts digital data into radio signals.
- A wireless Access Point (AP) connects wireless devices to the wired network.

Communication Example

- Laptop sends data wirelessly to the AP.
- AP forwards the data into the wired Ethernet network.
- Returning data follows the reverse path.

To Remember

- Wi-Fi = wireless connection to the LAN
- Access Point = bridge between wireless devices and the wired network



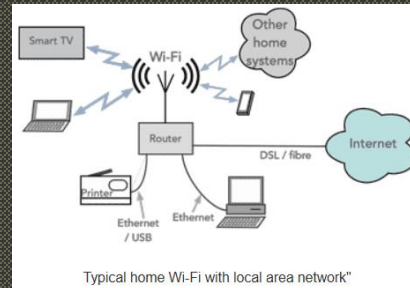
<https://www.electronics-notes.com/articles/connectivity/data-networks/lan-what-is-local-area-network-basics.php>

2.7 NETWORK MEDIA

Advantages of Wireless LANs

Advantages

- Mobility
 - Users can move while remaining connected.
- Flexibility
 - Devices can be added without installing cables.
- Lower installation effort
 - Useful where cabling is difficult.
- Supports many device types
 - Laptops
 - Smartphones
 - Tablets
 - Printers
 - IoT devices



Disadvantages

- Interference
 - Radio signals can be affected by other wireless devices and obstacles.
- Security risks
 - Wireless networks require proper authentication and encryption.
- Limited range
 - Signal strength decreases with distance and obstructions.
- Reduced performance with distance
 - Data rate and reliability may decrease farther from the AP.

Overall Comparison

- | | | |
|--|--|---|
| <ul style="list-style-type: none">• UTP copper<ul style="list-style-type: none">◦ Low cost◦ Easy to install◦ Good for short Ethernet links | <ul style="list-style-type: none">• Fiber optic<ul style="list-style-type: none">◦ High speed◦ Long distance◦ Excellent immunity to EMI◦ Very useful in industrial environments | <ul style="list-style-type: none">• Coaxial<ul style="list-style-type: none">◦ Mainly important today for cable TV and cable Internet• Wireless<ul style="list-style-type: none">◦ Flexible and mobile◦ No cable to the end device◦ More affected by interference, distance, and security concerns |
|--|--|---|

2.8 ETHERNET STANDARDS

Ethernet Standard Naming

- Ethernet standards are defined by the IEEE 802.3 family.
- Names such as 10BASE-T describe important characteristics of the Ethernet version
- In 1000BASE-T:
 - 1000 → 1000 Mb/s = 1 Gb/s
 - BASE → baseband signaling
 - T → twisted-pair copper

10BASE-T

- Speed:
 - 10 Mb/s
 - Uses twisted-pair copper cable.
 - Traditionally used two wire pairs.
 - Common in the 1990s.
 - Often used with hubs in older shared Ethernet networks.
 - Collisions could occur when hubs and half-duplex communication were used.
 - Today:
 - Mostly obsolete
- Mainly found in older or legacy systems

Ethernet Standards

Evolution of IEEE 802.3 Ethernet from 10 Mb/s to ultra-high-speed networks

Highly relevant to Industrial Networking Fundamentals:
Ethernet is the dominant wired LAN technology used with PLCs, HMIs, industrial PCs, switches, servers, drives, and automation devices.

Industrial Switch PLC HMI Servers / Data Center

1 Ethernet Standard Naming

10BASE-T

- 10 = 10 Mb/s
- BASE = baseband signaling
- T = twisted-pair copper

Ethernet standards are defined by the IEEE 802.3 family.

2 Core Standards Used in Practice

Standard	Speed	Medium	Notes
10BASE-T	10 Mb/s	Medium: twisted-pair copper	Traditionally used 2 wire pairs. Common in the 1990s. Often used with hubs / older shared Ethernet. Mostly obsolete today.
100BASE-TX	100 Mb/s	Medium: copper, Cat 5 or better	Uses 2 pairs. Still found in older buildings, older industrial equipment, and basic networks. Faster than 10BASE-T.
100BASE-FX	100 Mb/s	Medium: fiber-optic cable	Longer distances than copper. Excellent resistance to electrical noise and EMI. Useful for industrial environments and network backbones.

3 Gigabit and Multi-Gigabit Ethernet

Standard	Speed	Medium	Notes
1000BASE-T	1 Gb/s	Medium: Cat 5e or better copper	Uses all 4 wire pairs. Full-duplex. Typical maximum copper distance: 100 m. Very common in offices, schools, industrial networks, and LANs.
2.5GBASE-T / 5GBASE-T	2.5 Gb/s and 5 Gb/s	Medium: twisted-pair copper	Faster Ethernet over existing copper cabling. Useful for modern Wi-Fi access points, enterprise networks, and campus networks. Helps support devices needing more than 1 Gb/s.
10GBASE-T	10 Gb/s	Medium: twisted-pair copper	Higher-performance copper cabling. Common in servers, data centers, high-performance workstations, and network backbones. Full-duplex.

4 Fiber Ethernet Variants

Standard	Description
1000BASE-SX	Short-distance Gigabit Ethernet over fiber (typically up to 550 m)
1000BASE-LX	Longer Gigabit fiber links, campus/backbone use (up to ~10 km)
10GBASE-SR	Short-range fiber (typically up to 300-400 m)
10GBASE-LR	Long-range fiber (about 10 km class)
10GBASE-ER	Extended-range fiber (about 40 km class)

Industrial Importance:
Fiber Ethernet is especially valuable for long distances, high bandwidth, and environments with electrical noise / EMI, including links between buildings or plant areas.

5 High-Speed Ethernet for Data Centers

Standard	Speed	Notes
40G Ethernet	40 Gb/s	
100G Ethernet	100 Gb/s	High-performance data centers
200G / 400G		Very large modern data-center and AI networks
800G / 1.6T		

Main uses: data centers, cloud infrastructure, telecom networks, high-capacity backbones
Fiber is the main transmission medium
Short high-speed copper is used in some short-run applications

6 Why Ethernet Keeps Evolving

- Higher-speed Internet services**: Growing demand for faster access worldwide.
- Cloud computing**: More cloud services and applications.
- Video and streaming**: Higher resolution and more content.
- Wi-Fi improvements**: Faster Wi-Fi requires faster wired networks behind the scenes.
- Large data transfers**: Bigger files, backups, and distributed data.
- AI workloads**: AI and machine learning require high-speed, low-latency networks.
- Data-center growth**: Rapid expansion of data centers and cloud infrastructure.

7 Simple Ethernet Evolution

Standard	Speed
10BASE-T	10 Mb/s
100BASE-TX	100 Mb/s
1000BASE-T	1 Gb/s
2.5GBASE-T	2.5 Gb/s
5GBASE-T	5 Gb/s
10GBASE-T	10 Gb/s
40G	40 Gb/s
100G	100 Gb/s
200G / 400G	200 Gb/s / 400 Gb/s
800G / 1.6T	800 Gb/s / 1.6 Tb/s

8 Most Important for Industrial Networking

- ✓ 100BASE-TX – still common in PLCs, HMIs, drives, and older industrial equipment
- ✓ 1000BASE-T – very common for modern industrial Ethernet
- ✓ Fiber Ethernet – important for long distance and EMI immunity
- ✓ 10 Gigabit Ethernet – increasingly relevant for servers, industrial backbones, machine vision, and high-data applications
- ✓ 40G to 1.6T – useful mainly as industry awareness, not essential for introductory industrial networking

Quick Reference Summary

Standard	Medium	Speed	Notes
BASE-T	Ethernet over twisted-pair copper		
BASE-SX / LX / SR / LR / ER	Fiber Ethernet variants		
Higher number		Higher data rate	
Copper			Common for device connections
Fiber			Long distance, high speed, and excellent EMI immunity

ETHERNET BUILDS CONNECTIONS THAT MOVE THE WORLD

2.8 ETHERNET STANDARDS

100BASE-TX – Fast Ethernet

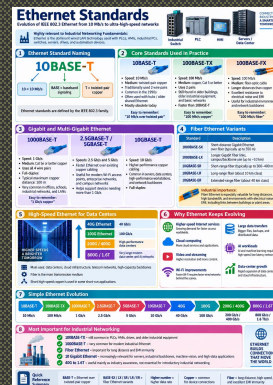
- Speed: 100 Mb/s
- Uses twisted-pair copper.
- Typically uses Cat 5 or better cable.
- Uses two wire pairs.
- Still found in:
 - Older buildings
 - Older industrial equipment
 - Basic networks
- Ten times faster than 10BASE-T.
- Largely replaced by Gigabit Ethernet in newer installations.

100BASE-FX – Fast Ethernet over Fiber

- Speed: 100 Mb/s
- Uses fiber-optic cable.
- Provides:
 - Longer transmission distances than copper
 - Excellent resistance to electrical noise and EMI
- Useful for:
 - Building-to-building connections
 - Industrial environments
 - Network backbones

1000BASE-T – Gigabit Ethernet

- Speed: 1 Gb/s
- Uses Cat 5e or better twisted-pair cable.
- Uses all four wire pairs.
- Normally operates full-duplex.
- Typical maximum copper distance:
 - 100 m
- Very common in:
 - Offices / Schools
 - Industrial networks
 - General-purpose LANs



Why Ethernet Speeds Continue to Increase

- Higher-speed Internet services
- Cloud computing
- Video and streaming
- Wi-Fi improvements
- Large data transfers
- AI workloads
- Data-center growth

2.8 ETHERNET STANDARDS

2.5GBASE-T and 5GBASE-T

- Speeds:
 - 2.5 Gb/s or 5 Gb/s
- Designed to provide faster Ethernet over existing copper cabling.
- Useful for:
 - Modern wireless access points
 - Enterprise networks
 - Campus networks
- Helps support devices requiring more than 1 Gb/s.

10GBASE-T

- Speed: 10 Gb/s
- Uses higher-performance copper cabling.
- Common in:
 - Servers
 - Data centers
 - High-performance workstations
 - Network backbones
- Operates full-duplex.

Fiber-Based Gigabit Ethernet

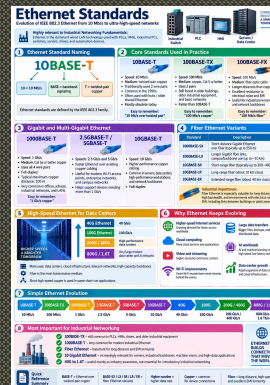
- 1000BASE-SX
 - Short-distance Gigabit Ethernet over fiber.
 - Common inside buildings and data centers.
- 1000BASE-LX
 - Designed for longer fiber links.
 - Used for campus and backbone connections.

10-Gigabit Fiber

- 10GBASE-SR
 - Short-range fiber.
- 10GBASE-LR
 - Long-range fiber.
 - Approximately 10 km class.
- 10GBASE-ER
 - Extended-range fiber.
 - Approximately 40 km class.

Industrial Importance

- Fiber Ethernet is especially valuable where:
 - Long distances are required.
 - High bandwidth is needed.
 - Electrical noise or EMI is present.
 - Connections run between buildings or plant areas.



2.8 ETHERNET STANDARDS

40 Gigabit and 100 Gigabit Ethernet

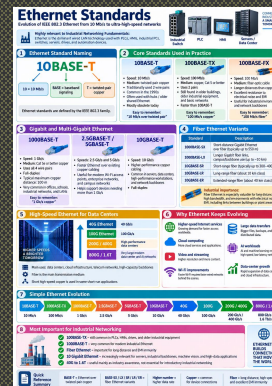
- Mainly used in:
 - Data centers
 - Cloud infrastructure
 - Telecom networks
 - High-capacity network backbones
- Fiber is the main transmission medium.
- Short high-speed copper connections may also be used in some applications.
- Typical roles:
 - 40G → high-capacity switch connections
 - 100G → data-center and core-network links

For Industrial Networking these standards are most in use:

- 100BASE-TX
 - Still common in PLCs, HMIs, drives, and older industrial equipment.
- 1000BASE-T
 - Very common for modern industrial Ethernet.
- Fiber Ethernet
 - Important for long distances and EMI immunity.
- 10 Gigabit Ethernet
 - Increasingly relevant for servers, industrial backbones, machine vision, and high-data applications.

200G and 400G Ethernet

- Designed for very high-capacity networks.
- Used by:
 - Large cloud providers
 - Telecom companies
 - Large data centers
 - AI and high-performance computing systems



800G and 1.6 Tb/s

- Represent the continuing evolution of Ethernet toward extremely high bandwidth.
- Intended mainly for:
 - Very large data centers
 - AI infrastructure
 - Cloud computing
 - High-performance backbone networks

2.9 Two Computers Communicating on a LAN

IP Address vs. MAC Address

Two Addresses Are Used

- A computer on a TCP/IP network uses:
 - IP address
 - MAC address

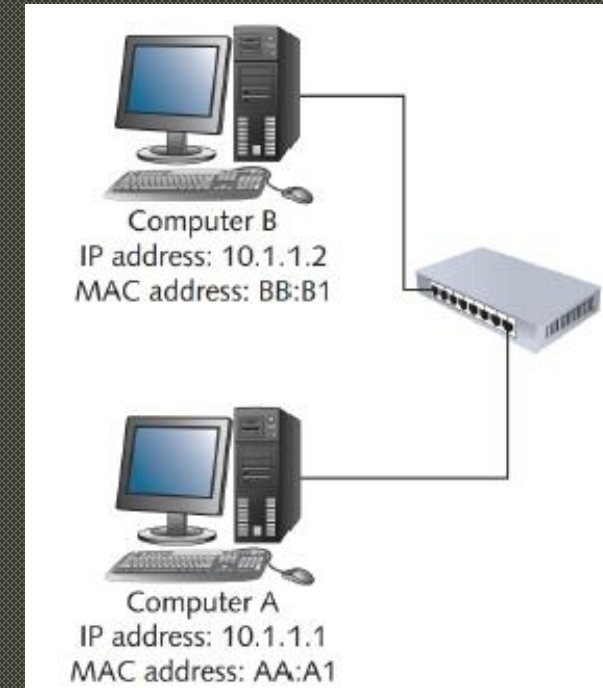
IP Address

- A logical Layer 3 address.
- Identifies the device and the network it belongs to.
- Used to determine where packets need to go.
- Example:
 - Computer A → 10.1.1.1
 - Computer B → 10.1.1.2

MAC Address

- A Layer 2 address associated with the network interface.
- Used for local Ethernet frame delivery.
- Example:
 - Computer A → AA:A1
 - Computer B → BB:B1

- Simple Analogy
 - IP address = postal address / location
 - MAC address = identifies the specific network interface used for local delivery
- IP addresses identify the communicating hosts at Layer 3.
- MAC addresses are used to deliver Ethernet frames on the local LAN.

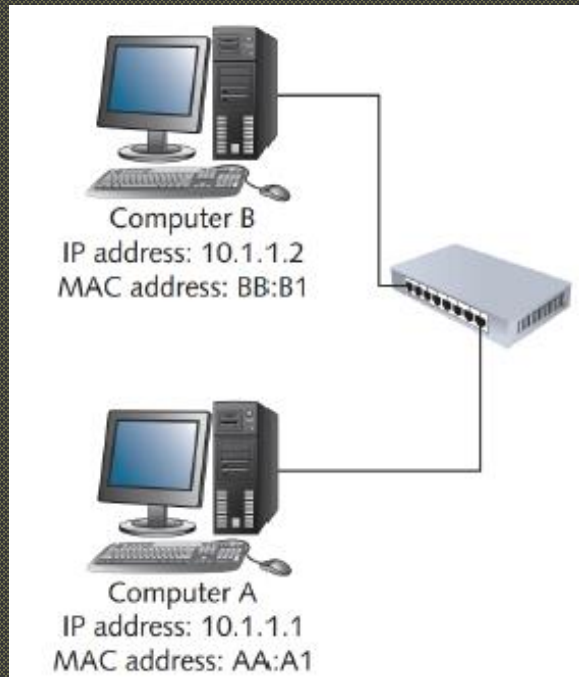


2.9 Two Computers Communicating on a LAN

Using Ping to Communicate

Example

- Computer A wants to determine whether Computer B is reachable.
- Computer A:
 - IP = 10.1.1.1
 - MAC = AA:A1
- Computer B:
 - IP = 10.1.1.2
 - MAC = BB:B1



Communication Process

- User enters:
 - ping 10.1.1.2
- Computer A creates an ICMP Echo Request.
- The IP layer adds:
 - Source IP address
 - Destination IP address
- Computer A needs Computer B's MAC address for local Ethernet delivery.
- If the MAC address is not already known:
 - ARP is used to discover the MAC address associated with 10.1.1.2.
- The NIC creates an Ethernet frame containing:
 - Source MAC address
 - Destination MAC address
- The frame is transmitted across the LAN.
- Computer B receives the frame and processes the ping request.
- Computer B sends an ICMP Echo Reply.
- Computer A receives the reply and confirms that Computer B is reachable.

Ping process on the same LAN:

Destination IP → ARP finds MAC → Ethernet frame → Ping request → Ping reply

2.9 Two Computers Communicating on a LAN

Using Names Instead of IP Addresses

- Users normally prefer names such as:
 - Server1
 - Laptop-123
- Networks still need an IP address to communicate.
- DNS – Domain Name System translates names into IP addresses.

Example

Instead of:

- ping 10.1.1.2

A user might enter:

- ping ComputerB

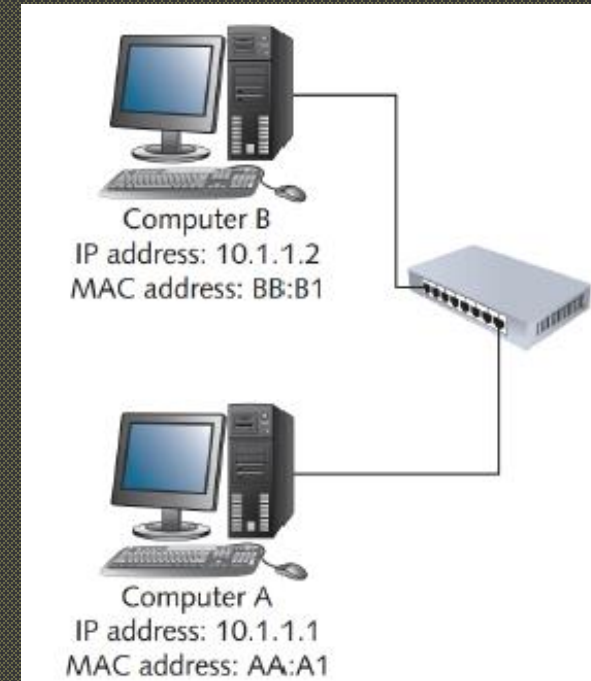
Then:

- DNS looks up ComputerB.
- DNS returns its IP address.

Example:

- ComputerB → 10.1.1.2

Communication then continues using the IP address.



- DNS = Name → IP address
- Example:
 - ComputerB → 10.1.1.2

2.9 Two Computers Communicating on a LAN

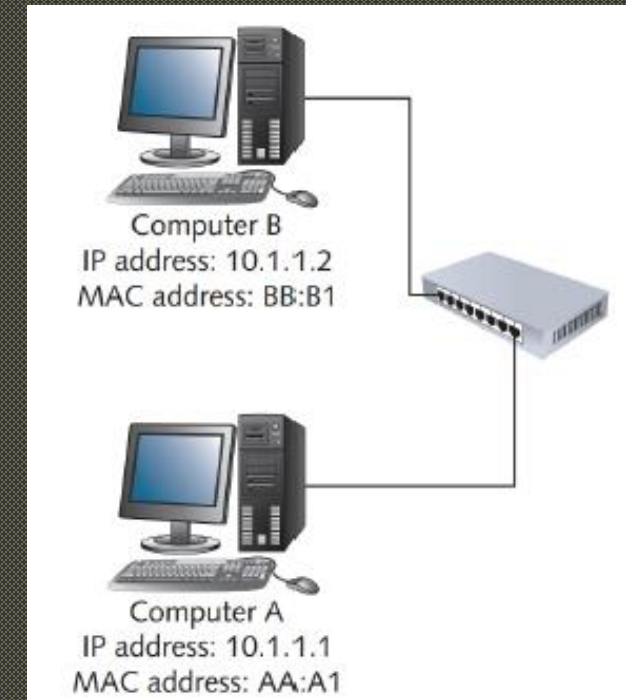
Complete Communication Process with DNS

If the user enters: ping ComputerB
the overall process is:

1. Name resolution
 - DNS finds Computer B's IP address.
2. Create ping request
 - Computer A creates an ICMP Echo Request.
3. Add IP addresses
 - Source IP = Computer A
 - Destination IP = Computer B
4. Determine local MAC address
 - ARP finds Computer B's MAC address if needed.
5. Create Ethernet frame
 - Source MAC = Computer A
 - Destination MAC = Computer B
6. Transmit across the LAN
 - The Ethernet frame travels through the local network.
7. Computer B receives the request
 - B processes the frame, packet, and ICMP message.
8. Computer B replies
 - B sends an ICMP Echo Reply back to Computer A.

Complete Simple Flow
Computer name

- ↓
- 1. DNS finds destination IP address
- ↓
- 2. ICMP Echo Request created
- ↓
- 3. IP header added → IP packet created
- ↓
- 4. ARP finds destination MAC address, if needed
- ↓
- 5. Ethernet frame created
- ↓
- 6. Frame sent across LAN
- ↓
- 7. Destination processes the request
- ↓
- 8. ICMP Echo Reply returned



References

- Computer Networks and Internets, Douglas e. Comer (9-15), (241-252)

INDUSTRIAL NETWORKING

MENG 4010

Thank you!

Discussions?